# Addressing incomplete soil sample data in Soil Enrichment Protocol projects

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**Abstract**

This document provides guidance for statistical techniques to reduce bias from missing samples in projects that estimate carbon offsets using the Climate Action Reserve's Soil Enrichment Protocol (SEP). Future versions of this guidance may generalize and become a separate guidance document housed at the Climate Action Reserve for the SEP (and potentially for other methodologies). The immediate plan, however, is for Indigo Ag to attach this document (and its review) to its first monitoring plan associated with its SEP project, expected later in 2021. Another appendix of that monitoring plan will document Indigo's use of one or more techniques described here, following the guidance in this document for the application of the technique and for the documentation of its use. That way, verifiers of Indigo's SEP project can consult this document to confirm that Indigo is following best statistical practice.

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# 1 Introduction

## 1.1 Context

The Climate Action Reserve (Reserve) Soil Enrichment Protocol (SEP) provides guidance on how to account for, report, and verify greenhouse gas (GHG) emission reductions and removals on agricultural lands due to the adoption of sustainable land management practices [1]. GHG emission

reductions and removals covered by the SEP include $CO_2$, $N_2O$, and $CH_4$ emission reductions as well as soil organic carbon (SOC) stock change.

While the emission reductions of some sustainable land management practices can be calculated with default equations (e.g., reduced $CO_2$ and $N_2O$ emissions due to reduced biomass burning), others are predicted with biogeochemical models (e.g., increased SOC stocks due to adoption of cover crops and no-till practices). Soil attributes at the project initiation—such as carbon concentration (%SOC), pH, texture, and bulk density—are key inputs to these biogeochemical models. For this reason, soil attributes must be directly measured at the initiation of an SEP project, as well as at least every five years thereafter [1, Section 5]. These directly measured soil attributes are used as input to biogeochemical models, which are then used to predict GHG emissions and SOC levels under two scenarios:

- Baseline scenario: Assuming growers had not adopted new sustainable land management practices
- Project scenario: Accounting for the additional sustainable land management practices that growers adopted due to the project[1]

The GHG emission reductions and removals are then estimated as the difference in emissions and SOC stock between the project and baseline scenarios.

As described in Table 6.2 of the SEP, the soil measurements must be taken at randomly selected locations (points) according to a sample design with the aim of collecting a representative sample of soils. The biogeochemical models make projections under the above scenarios at each sampled point. However, it may not be possible to collect and analyze soil from all selected points due either to reasons that may be associated with soil attributes (e.g., frozen soil or localized flooding), or reasons that are likely unassociated with soil attributes (e.g., equipment failure, lost sample labels, laboratory mishap). These missing points represent survey nonresponse. As a first step to addressing survey nonresponse, all possible attempts should be made to collect data from the selected points in follow-up efforts. However, due to resource and logistical constraints, physical barriers (e.g., irrigation equipment below the surface at selected points), or other challenges, some samples may remain missing despite best efforts to collect and analyze them. The missing points could represent entire fields or portions of fields.

If the nonresponse is due to reasons unrelated to the soil attributes used as inputs to the biogeochemical models (and thus independent of estimated GHG emission reductions and removals), then the nonresponse would not cause bias. However, if nonresponse is correlated with emission reductions and removals, then a naive analysis of the observed data could lead to biased estimates because points with high or low emission reductions might be over- or under-represented in the sample [2, 3].

The distinction of when nonresponse does and does not cause bias is related to common classifications of missing data. In particular, when soil samples are missing completely at random (MCAR) [4], then by definition nonresponse cannot be related to field attributes and there is no concern for bias. In contrast, if the soil samples are missing at random (MAR) or missing not at random (MNAR) [4], then nonresponse could cause bias if the missingness is associated with emission reductions or removals. Said another way, MCAR is a sufficient but not necessary condition for ensuring a lack of bias in the presence of nonresponse; a lack of association between missingness and the outcome is a necessary condition.

[1] Please see Section 3.4 of the SEP for details on additionality.

The remainder of this document provides guidance on analytic approaches for addressing missing soil samples in SEP projects, with the aim of mitigating potential bias and reducing variance. Before making any analytic adjustments to address missing data, all reasonable attempts should be made to collect and analyze samples at all selected points.

### 1.2 Balancing completeness, timeliness, and uncertainty

In large SEP projects comprising thousands of growers and many times that many fields (called "aggregated projects"), completing a sample may take years because opportunities for sampling are brief and spaced months apart. For example, a field may be available for sampling during a two-week period after harvest of a cash crop and before the subsequent cover crop is too tall for sampling crews to navigate it; if one misses that window, one must wait six months for another opportunity to sample the field. Insisting on a complete sample jeopardizes the ability of such projects to issue credits in a timely manner. Meanwhile, insisting on smaller sample sizes (that are more likely to be completed) results in intolerably high uncertainty. We think the optimum lies between these extremes: choose a sample size that is large yet plausible to complete, and use the techniques described here to minimize bias from a relatively small fraction of missing data.

### 1.3 Background on soil samples and sample points in SEP projects

In SEP projects, a sample design selects *points* at which to take multiple *samples* of soil for various measurements. The SEP requires that project developers measure the concentration of soil organic carbon (abbreviated %SOC) and bulk density in their project [1, Section 6.5]. Other soil measurements, such as texture and pH, can be taken if desired (for example, to improve accuracy of biogeochemical model predictions).

Different equipment is used for these measurements. As a result, some or all samples may be missing at a point. For example, bulk density cores are particularly prone to breaking due to the amount of force applied, resulting in some points having a %SOC measurement but missing a bulk density measurement.

In this document, our main focus is on the missingness of %SOC measurements (because they vary significantly in space and are a key variable that SEP projects aim to influence). Thus, we say that a *sample point is missing* if its %SOC measurement is missing. Because of the importance of %SOC measurements, we recommend that it be measured at every point chosen by the sample design.

This document also covers missingness of bulk density, texture, and pH measurements. SEP projects may take these measurements at a random subset of points where %SOC is measured because the former measurements are expensive to take, lead to more injuries to soil samplers (in the case of bulk density), and/or vary less within fields than %SOC does [5]. At points where these measurements were missing—either deliberately so (to reduce costs) or due to unforeseen challenges (e.g., sample lost in shipment; part of a field was flooded)—they can be imputed using models, subject to the requirements in Section 5.3.

## 2 Overview

Figure 1 shows the overall steps and decisions for addressing incomplete soil samples after all reasonable attempts have been made to collect and analyze missing samples. The process in Figure

1 begins with a description of why samples were missing and an assessment of whether the missing samples could potentially cause bias (as described in more detail in Section 3). If these investigations conclude that there is potential for bias, then statistical adjustments must be made at either the field or sub-field level (see Section 4) with approved statistical methods (see Sections 5 and 6) to minimize potential bias. In addition to the adjusted estimates of GHG emission reductions/removals, an unadjusted analysis (see Sections 5.1 and 6.2.1) must be reported for comparison.

If the investigations outlined in Section 3 do not raise concerns for potential bias, then only an unadjusted analysis is required (the variance will also only reflect the observed sample size; see, e.g., Equation (4)). However, a statistical adjustment method may also be conducted with the aim of reducing variance (see Section 3.3).

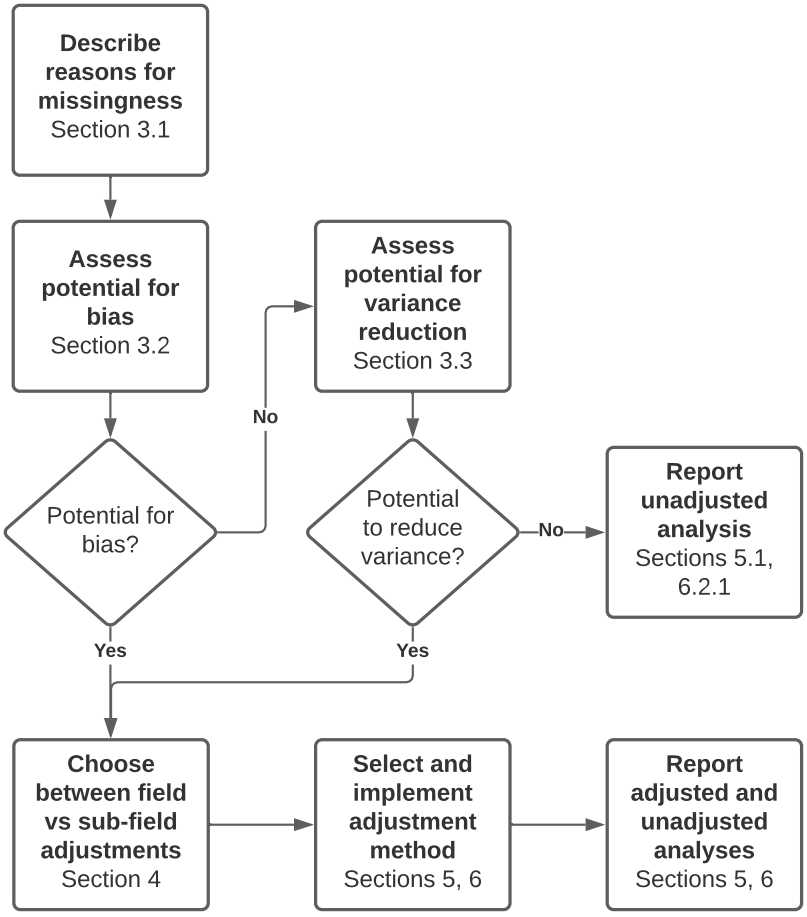



Figure 1: Overall steps and decisions in addressing incomplete soil samples

# 3 Investigating missing soil samples

## 3.1 Describing reasons for missing soil samples

The reason that each planned soil sample is missing (for all types of samples: %SOC, bulk density, pH, and texture) should be tabulated and categorized as to whether or not it is related to field attributes associated with emission reductions and removals, and thus a potential concern for bias. Table 1 provides an example, though the most relevant categorization of reasons for missingness may vary across SEP projects. However, in all cases the reasons should be grouped into those that are potentially concerning (related to field attributes) and those that are not concerning (unrelated to field attributes). In rare cases, it may not be known why a sample is missing, in which case the reason should be categorized as "Unknown."

Table 1: Reasons why planned soil samples are missing (example template)

| | Count (% of total planned samples) | | |
|---|---|---|---|
| Reason for missingness | %SOC | Bulk density | Texture and pH |
| Potentially concerning (related to field attributes) | | | |
| Seasonal delay | $n$ (%) | $n$ (%) | $n$ (%) |
| Short-term weather delay | $n$ (%) | $n$ (%) | $n$ (%) |
| Crop already established | $n$ (%) | $n$ (%) | $n$ (%) |
| Livestock in area temporarily | $n$ (%) | $n$ (%) | $n$ (%) |
| Unknown | $n$ (%) | $n$ (%) | $n$ (%) |
| Not concerning (unrelated to field attributes) | | | |
| Field equipment failure (unrelated to the soil) | $n$ (%) | $n$ (%) | $n$ (%) |
| Inadvertently missed by field staff | $n$ (%) | $n$ (%) | $n$ (%) |
| Laboratory error | $n$ (%) | $n$ (%) | $n$ (%) |

In addition, details should be provided for each missing soil sample, including:

1. The date at which the problem occurred or was reported
2. A 1–2 sentence description of the problem, and
3. An overall categorization to cross-reference the reasons for missingness in Table 1.

As an example, the 1–2 sentence descriptions might be along the lines of "Sample tested positive for excess lime, but results were reported as total carbon percent. Lab could not guarantee that samples received acid pre-treatment," or "Sample could not be taken due to localized flooding."

Verifiers are encouraged to use the information in Table 1, and the more detailed information about individual soil samples, to understand the causes of missing soil measurements and to better interpret the adjustments made to address them.

## 3.2 Assessing implications for bias

To assess whether the potentially concerning missing %SOC samples might be associated with GHG emission reductions and removals, the marginal distributions of factors associated with emission

reductions and removals should be compared between i) all points in the planned sampled, and ii) points with observed %SOC, such as in Table 2. Table 2 is an example, and the exact attributes and categorizations of attributes may vary across SEP projects. Please see Section 5.2.1 below for additional guidance on attributes that may be appropriate to include in Table 2.

Table 2: Attributes of i) all points in the planned sample versus ii) points with observed %SOC. For categorical variables such as the binned total precipitation, the $n$ values in a column sum to the number of such samples. For continuous variables such as SSURGO %SOC, the mean and standard deviation are provided.

| Attribute | All ($n$) | Observed ($n$) | Standardized difference |
|---|---|---|---|
| Land Resource Region | | | $d$ |
| F | $n$ (%) | $n$ (%) | |
| K | $n$ (%) | $n$ (%) | |
| M | $n$ (%) | $n$ (%) | |
| ⋮ | ⋮ | ⋮ | ⋮ |
| Total precipitation | | | $d$ |
| $<$ 10 inches | $n$ (%) | $n$ (%) | |
| 10–15 inches | $n$ (%) | $n$ (%) | |
| 15–20 inches | $n$ (%) | $n$ (%) | |
| ⋮ | ⋮ | ⋮ | ⋮ |
| SSURGO clay content | | | $d$ |
| $<$ 10% | $n$ (%) | $n$ (%) | |
| 10–15% | $n$ (%) | $n$ (%) | |
| 15–20% | $n$ (%) | $n$ (%) | |
| ⋮ | ⋮ | ⋮ | ⋮ |
| SSURGO %SOC | mean (sd) | mean (sd) | $d$ |
| ⋮ | ⋮ | ⋮ | ⋮ |
| Practice management change | | | $d$ |
| Conventional tillage to no-till | $n$ (%) | $n$ (%) | |
| Conventional tillage to reduced tillage | $n$ (%) | $n$ (%) | |
| Began manure application | $n$ (%) | $n$ (%) | |
| Began returning residue | $n$ (%) | $n$ (%) | |
| Corn/soy rotation to corn/wheat | $n$ (%) | $n$ (%) | |
| ⋮ | ⋮ | ⋮ | ⋮ |

The standardized difference [6] between i) all points in the planned sample and ii) points with observed %SOC should be calculated for all attributes in Table 2. For example, let $p_{\text{all},k\ell}$ and $p_{\text{obs},k\ell}$ be the proportion of points in category $\ell$ for attribute $k$ among all points in the planned sample and points with observed %SOC, respectively. The proportions sum to 1 within an attribute ($\sum_\ell p_{\text{all},k\ell} = 1$ and $\sum_\ell p_{\text{obs},k\ell} = 1$). We define the standardized difference for a dichotomous

attribute as[2]

$$d_k = \frac{p_{\text{obs},k2} - p_{\text{all},k2}}{\sqrt{p_{\text{all},k2}(1 - p_{\text{all},k2})}} \tag{1}$$

which scales the difference in proportions by the standard deviation of the planned sample.[3] The generalization of (1) to attributes with more than 2 categories is given in Appendix A. The standardized difference for continues variables with means $\bar{x}_{\text{all},k}$ and $\bar{x}_{\text{obs},k}$ is

$$d_k = (\bar{x}_{\text{obs},k} - \bar{x}_{\text{all},k})/s_{\text{all}} \tag{2}$$

where $s_{\text{all}}$ is the standard deviation among all points (the planned sample). Equation (2) should be used for any continuous variables in Table 2 that are not categorized.

If $|d_k| \leq 0.1$ for all attributes $k$, then there may be minimal or no concern for bias. The threshold of 0.1 is commonly used in observational medical and behavioral studies [8, 9], though a threshold of 0.25 is also used in some scientific fields [10], and future work may indicate that a value other than 0.1 is more appropriate for SEP projects. Statistical hypothesis tests and $p$-values should not be used to assess whether there is a concern for bias, because $p$-values are a function of sample size (e.g., for large sample sizes, the point null hypothesis of no difference may be rejected even if the magnitude of the difference is too small to cause a concern for bias).

The standardized differences should be recomputed after making adjustments to assess how well the adjustment reduced concern for potential bias. For example, if post-stratification (Section 5.2) is used, the standardized differences should be recomputed within each post-stratum. If the post-stratified standardized differences are not all less than 0.1 in magnitude, then changes can be made to the post-stratification (e.g., splitting post-strata into multiple post-strata) to further reduce imbalance, though the post-strata sample size requirements in Section 5.2.2 must still be met. As long as the decision on whether to change the post-strata are based on the standardized differences and not on estimated GHG emission reductions/removals, there is no concern for cherry-picking results. This recommendation is in line with common practice in observational research for selecting matches that achieve covariate balance [11].

**Note 1: Limitations and context for interpreting the standardized difference**

The standardized difference $d_k$ assesses differences in marginal distributions, so there may be differences in joint distributions that go undetected. In addition, for attributes with many categories, the counts in each category may be small (e.g., less than 10 sample points) leading to unstable estimates.

The limitations and context noted above, together with the reasons for missingness in Table 1, should be taken into account when assessing the standardized differences and determining whether

[2] As noted above, In SEP projects the relevant difference is between i) all points in the planned sample and ii) points with observed %SOC; the set of all points is a probability sample from the target population, but emission reductions can only be estimated at points with observed %SOC. Therefore, we must assess whether the emission reductions for points in group (ii) could be biased for the emission reductions for points in group (i). This is in contrast to the setting of [6], in which the relevant comparison would be between points with observed versus missing %SOC. The comparison between groups (i) and (ii) is in keeping with typical nonresponse bias analyses, such as [7].

[3] Note that for dichotomous variables, the absolute value of the standardized difference between the proportions in category $\ell = 1$ is identical to that between the proportions in category $\ell = 2$ because $p_{\text{all},k1} = 1 - p_{\text{all},k2}$ and $p_{\text{obs},k1} = 1 - p_{\text{obs},k2}$. Therefore only one standardized difference $d$ is reported per attribute.

the concerning missing soil samples could potentially cause bias. The reasoning and conclusion must be documented and address all of the points above.

Other descriptive summaries that may be considered but which are not required include:

- A choropleth map showing the location of all fields in the project, with coloring or shading to indicate the proportion of sample points that are missing
- A histogram showing the distribution of sampling proportions within fields
- A scatter plot comparing field acreage and proportion sampled

### 3.3 Assessing potential for variance reduction

If there is minimal to no potential for bias, then the analysis can focus on reducing variance. In particular, post-stratification can reduce variance if GHG emission reductions/removals are homogeneous within post-strata (see Section 5.2). To assess the potential for using post-stratification to reduce variance, the overall distribution of outcomes can be visually inspected, such as with a histogram for each GHG emission reduction/removal. A very tight distribution may indicate lower potential for identifying post-strata with greater within-stratum homogeneity than in the full sample. Conversely, a very spread out distribution may indicate greater potential to identify post-strata with greater within-stratum homogeneity, though this is not guaranteed. In addition to visually inspecting outcome distributions, summary statistics such as the standard deviation can be calculated overall and within groups. All visuals and statistics should be discussed with agronomists, soil scientists, and other domain experts. The rationale for any decisions related to the potential to reduce variance, as well as all visuals and statistics used to support this decision, must be documented.

**Note 2: Constraints on reducing variance through post-stratification**

Section 5.2.2 provides guidelines when post-stratifying for variance reduction to prevent post-strata from being too small or too sparsely sampled.

## 4 Determining whether to make adjustments at the field or sub-field level

The statistical adjustment methods in Section 5 can be applied at both the field and sub-field levels. For example, post-stratification could group together similar fields or similar sub-field areas. In the latter approach, the sub-field areas grouped together in a single post-stratum would not necessarily belong to the same field. For example, if two nearby fields both have a portion that is predominantly loam and another portion that is sandy loam, then post-stratification at the sub-field level might combine the loam portions of each field into one post-stratum and the sandy loam portions of each field into another post-stratum. While not currently recommended, calibration and multilevel modeling with post-stratification can also be implemented at both the field and sub-field levels (see Appendix F).

Table 3 provides a list of topics and considerations relevant for deciding between field level and sub-field level nonresponse adjustments. For a particular SEP project, the topics and considerations

in Table 3 should be taken into account when selecting an approach. Table 3 is not necessarily exhaustive, and there may be other relevant topics and considerations for any given SEP project.

Table 3: Topics and considerations for deciding between nonresponse adjustments at the field versus sub-field level

| **Topic** | **Considerations** |
|---|---|
| Geographic boundaries | It may be more challenging to identify reliable boundaries for sub-field areas than for entire fields. |
| Within-field variation in soil attributes | If fields are heterogeneous with respect to soil attributes such as %SOC, bulk density, soil texture, and pH, then making adjustments at the sub-field level will lead to more homogeneous groups, which can result in greater variance reduction for some methods such as post-stratification. |
| Variance estimation | In sampling designs, it is typical for fields to be the primary sampling unit. In these sampling designs, the estimation of variance may be more complex when forming post-strata at the sub-field level. However, bootstrap or jackknife methods may be used in these situations where resampling is done at the primary unit level (see Examples 4 and 10) |
| Computational demand | It may be less computationally intensive to adjust for nonresponse at the field level, both for the reason noted above regarding the use of bootstrap and jackknife methods, and because field-level data may be represented with fewer rows. |

### 4.1 Addressing missingness in partially sampled fields if post-stratifying at the field level

Some fields are *partially sampled*, meaning that some of the SOC concentration measurements are taken, but others are missing. When post-stratifying at the field level in sampling designs for which fields are the primary sampling units, one should check whether missingness within fields is associated with emission reductions/removals. This association is hard to assess given the small number of points within field. (Ideally, a Table 2 would be made for each field.) Here, we give a workaround, depicted in the flow diagram in Figure 2.

The first step is to pool information among partially-sampled fields to compute new standardized differences, denoted as $d_k^{\text{partial}}$, using just the partially-sampled fields. If in the partially-sampled fields $|d_k^{\text{partial}}| \leq 0.1$ for all attributes $k$ and categories $\ell$, then field-level post-stratification can proceed without further adjustments because the estimated averages in each field appear to be unbiased.[4]

[4]Combining points across all partially sampled fields (separately for observed and missing points) could potentially mask within-field differences. For example, one field might be missing points due to localized flooding in a lowland portion, and another field might be missing points due to cattle grazing in a highland portion. In aggregate across the two fields, the proportion of missing points in lowland areas might seem similar to the proportion of observed points in lowland areas, but within each field the proportions might not be similar. This should be considered as a potential limitation when interpreting results.

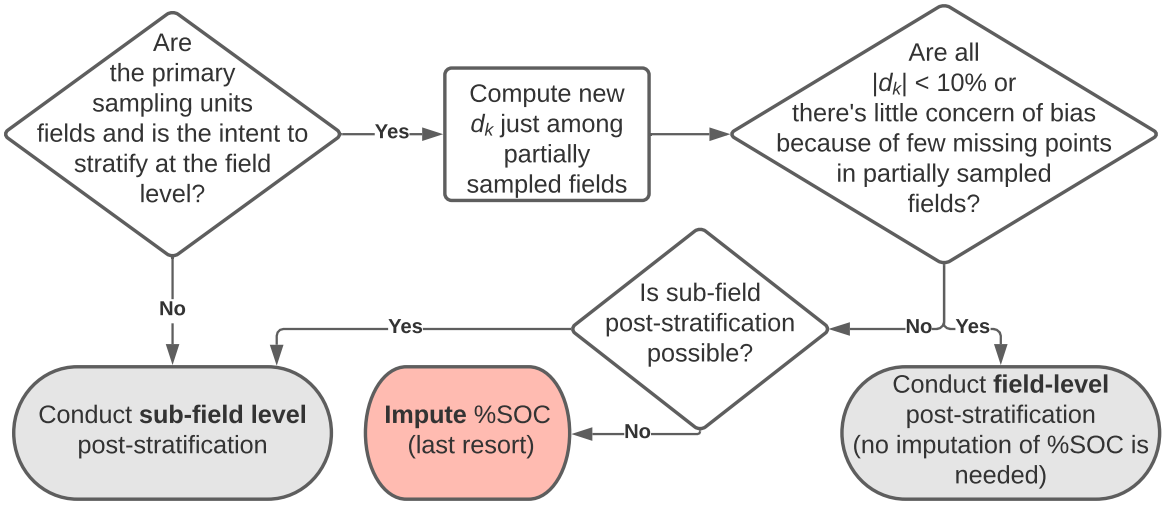



Figure 2: Flow diagram for deciding whether to impute SOC concentration values to handle bias in partially-sampled fields when post-stratifying at the field level for sampling designs in which fields are the primary units. This imputation is a last resort to be used only when sub-field post-stratification is infeasible (recall the considerations in Table 3).

If, however, $|d_k^{\text{partial}}|$ are not all less than 10%, then simply post-stratifying at the field level may not suffice to eliminate concern for bias. In this case, it is recommended to post-stratify at the sub-field level (see, e.g., Examples 4 and 10). If, however, post-stratifying at the sub-field level is infeasible (e.g., because it is difficult to obtain data that vary at sub-field levels), then one may, as a last resort, impute SOC concentration values according to the guidelines described in Section 5.3.

The decision of whether to make adjustments at the field versus sub-field level must be documented in a written report and address, at a minimum, all of the topics and considerations in Table 3.

# 5 Selecting and implementing a statistical adjustment method

This section describes statistical methods that can be used to estimate total GHG emission reductions/removals and variance about the total. All examples are presented in terms of totals, but it is straightforward to alter the formulas to estimate areal means and associated variances. The decision and rationale for the selected method should be documented.

The estimators in Examples 1 through 5 of this section are applicable when GHG emission reductions/removals are predicted with biogeochemical models for which calibration is not necessary. This assumption simplifies the calculations and presentation, though in practice biogeochemical models typically need to go through a calibration process before being used in an SEP project.[5]

[5]See the *Requirements and Guidance for Model Calibration, Validation, Uncertainty, and Verification* appendix

While necessary for minimizing bias, the calibration process can introduce additional uncertainty that must be accounted for. Please see Section 6 for examples of how to extend the estimators in the current section to SEP projects that use Bayesian-calibrated biogeochemical models.

### 5.1 Unadjusted analysis

As described in section 2, an unadjusted analysis is always required for comparison to adjusted analyses. An unadjusted analysis may also be provided as the sole analysis if there is no concern over potential nonresponse bias as discussed in Section 3.2.

In an unadjusted analysis, the observed data are analyzed as though there were no missing data. The exact calculations will depend on the sampling design, though we provide examples below.

**Example 1: Unadjusted analysis of a two-stage sampling design where fields are selected with probability proportional to size (PPSWR) and points are selected with a simple random sample with replacement (SRSWR)**

This is the same sampling design as in Appendix D.4 of the SEP [1]. Let $y_{ij}$ be the GHG emission reduction/removal at point $j$ in field $i = 1, \ldots, N$. In the first stage, $n^{\text{plan}}$ field selections are made with PPSWR, and in the second stage an SRSWR of $m^{\text{plan}}$ points are drawn from each field selection. If field $i$ is selected $k_i$ times in the first stage, there would be $k_i \times m^{\text{plan}}$ points from field $i$ in the sample. If no data were missing, there would be $n^{\text{plan}} \times m^{\text{plan}}$ samples.

Let $n$ be the observed number of field selections with partial or full sampling information; let $m_i$ be the observed number of samples in field selection $i$; and let $\bar{y}_i = m_i^{-1} \sum_{j=1}^{m_i} y_{ij}$ be the observed mean in field selection $i$. Also, let $A_i$ be the area of field $i$, and let $A = \sum_{i=1}^{N} A_i$ be the total area of the project. Because the estimated total in each field is $A_i \bar{y}_i$ and at each draw field $i$ is selected with probability $p_i = A_i/A$, the Hansen-Hurwitz estimator of the total is

$$\hat{\tau} = \frac{1}{n} \sum_{i=1}^{n} \frac{A_i}{p_i} \bar{y}_i = \frac{A}{n} \sum_{i=1}^{n} \bar{y}_i = \sum_{i=1}^{n} \sum_{j=1}^{m_i} w_i y_{ij} \tag{3}$$

where the weights are $w_i = A/(nm_i)$. The estimator of variance is

$$\widehat{\text{Var}}(\hat{\tau}) = \frac{A^2}{n} s^2 \tag{4}$$

where

$$s^2 = \frac{1}{n-1} \sum_{i=1}^{n} (\bar{y}_i - \hat{\mu})^2 \tag{5}$$

and $\hat{\mu} = \hat{\tau}/A$ is the Hansen-Hurwitz estimator of the areal mean. To estimate the variance of the areal mean $\hat{\mu}$, the variance (4) would be divided by $A^2$.

to the SEP [1]

Example 2: Extension of Example 1 when sampling is proportional to estimated size

Building on Example 1, now suppose that in the first stage of the design, sampling was done with probability proportional to *estimated areas* with replacement. This situation can arise, for example, when field boundaries are corrected after randomization due to on-site observations. Let $\hat{A}_i$, $i = 1, \ldots, N$ be the estimated areas of fields, i.e., the known areas of fields at the time of randomization based on their digital boundaries at the time; these $\hat{A}_i$ are estimates of the fields' final areas $A_i$ after corrections are made. Let $\hat{A} = \sum_{i=1}^{N} \hat{A}_i$ be the total estimated area. In this design, the sampling probabilities are $p_i = \hat{A}_i / \hat{A}$, which leads to the following estimator for the total [12, Sec. 11.9]:

$$\hat{\tau} = \frac{1}{n} \sum_{i=1}^{n} \frac{A_i}{p_i} \bar{y}_i = \frac{\hat{A}}{n} \sum_{i=1}^{n} \frac{A_i}{\hat{A}_i} \bar{y}_i = \sum_{i=1}^{n} \sum_{j=1}^{m_i} w_i y_{ij} \tag{6}$$

where the weights are $w_i = (\hat{A} A_i)/(\hat{A}_i n m_i)$.
The estimator of variance is

$$\widehat{\mathrm{Var}}(\hat{\tau}) = \frac{\hat{A}^2}{n} s^2 \tag{7}$$

where

$$s^2 = \frac{1}{n-1} \sum_{i=1}^{n} \left( \frac{A_i}{\hat{A}_i} \bar{y}_i - \hat{\mu} \right)^2 . \tag{8}$$

and $\hat{\mu} = \hat{\tau}/A$ is the Hansen-Hurwitz estimator of the areal mean. As in Example 1, to estimate the variance of the areal mean $\hat{\mu}$, the variance (4) would be divided by $A^2$.
As expected, Equations (6), (7), and (8) simplify to Equations (3), (4), and (5), respectively, when $\hat{A}_i = A_i$ for all $i = 1, \ldots, N$.

### 5.2 Post-stratification

Post-stratification uses auxiliary variables to group units (either fields or sub-field areas) into mutually exclusive strata, or cells, after the sample has been designed [13, 14]. For example, one post-stratum might consist of all fields in Land Resource Region 'M' that: i) are predominantly loam, and ii) changed from conventional tillage to no-till. Mean emission reductions and removals are estimated within each post-stratum separately. Each post-stratum's mean estimate is multiplied by the post-stratum's area to estimate the post-stratum total emission reductions and removals, and these post-stratum totals are then summed across all post-strata to estimate the total emission reductions for the project.

Post-stratification is closely related to nonresponse weighting. Whereas post-stratification reweights the observed soil samples to be representative of the population (here, all land in the project), nonresponse weighting reweights the observed soil samples to be representative of the planned sample. When the sample is representative of the population, these two weighting approaches will lead to nearly identical results, though the population quantities are of primary interest.[6]

[6] In some contexts, the term *post-stratification* is reserved for situations where there are no missing data and the term *population-based adjustment cell weighting* [15] or *population weighting adjustment* [16, p. 69] is used in the

### 5.2.1 Post-stratification variables

Appropriately selecting the post-stratification variables is critical for achieving the goals of minimizing potential nonresponse bias and reducing variance. Nonresponse bias can be removed by forming post-strata within which emission reductions/removals are independent of the propensity for soil samples to be missing [17]. In particular, these goals can be achieved by forming post-strata within which one or both of the following hold: i) GHG emission reductions/removals are homogeneous, and/or ii) the probabilities of response are homogeneous [3]. While both approaches control non-response bias, post-strata that are homogeneous with respect to outcomes also reduce variance [13, 3], and there is concern that post-strata that are homogeneous with respect to response probabilities but not emission reductions/removals can increase variance [18]. This insight leads to the following key points:

**Note 3: Key points on selecting post-stratification variables**

To minimize nonresponse bias and variance, post-stratification variables should be highly associated with GHG emission reductions/removals. If possible, the post-stratification variables should also be highly associated with response probabilities. These variables may differ across different types of emission reductions/removals due to differences in the underlying biogeochemical processes.

Based on the key points above, Table 4 provides example variables that may be important to consider for SOC emission reductions/removals. Other variables may be relevant for other types of emission reductions/removals. In practice, continuous variables such as total annual precipitation may need to be categorized before using them to form post-strata. Table 4 is not necessarily exhaustive. For any given SEP project, the selection of the most relevant variables, as well as the most appropriate categorization of continuous variables, should be chosen through discussions with agronomists, soil scientists, and other domain experts, and the rationale must be fully documented.

presence of nonresponse. However, the calculations are the same in both situations.

Table 4: Examples of candidate post-stratification variables for SOC emission reductions/removals

| Variable |
|---|
| Region |
| Land Resource Region |
| Weather (e.g., from NLDAS[7]) |
| Total precipitation during reporting period |
| Average temperature during reporting period |
| Soil attributes (e.g., from SSURGO [19] or POLARIS [20]) |
| SOC concentration |
| Texture, particularly clay content |
| Change in management practice |
| Conventional tillage to no-till |
| Conventional tillage to reduced tillage |
| Began manure application |
| Began retaining residue |
| Corn/soy rotation to corn/wheat |

#### 5.2.2 Post-strata sample size

While post-stratification can reduce variance by forming homogeneous groups, post-strata that are too small can increase variance [14]. There are different recommendations on minimum post-strata sample size. For example, for general surveys Cochran [12] recommends more than 20 observations in each post-stratum, and for forestry studies Westfall et al. [21] recommends at least 25 within each post-stratum to avoid bias in the standard error in proportion forestland analyses. Based on these recommendations:

- SEP projects that employ a sampling design in which points are the primary sampling units should have at least 25 point selections in each post-stratum.
- SEP projects that employ a sampling design in which fields are the primary sampling units and which post-stratify at the field level should have a minimum of 25 field selections in each post-stratum.
- SEP projects that employ a sampling design in which fields are the primary sampling units and which post-stratify at the sub-field level must:
  - Have at least 25 point selections in each post-stratum, and
  - Have a sufficient number of fields appearing in each post-stratum so that no more than 30% of bootstrap resamples are rejected (see Examples 4 and 10).

If the above criteria are not met, then the post-strata should be changed, such as by collapsing two post-strata into one as described in Section 5.2.3. Future work may indicate that there is a different minimum sample size more appropriate for SEP projects, at which point this guidance

[7]North American Land Data Assimilation System (NLDAS): https://ldas.gsfc.nasa.gov/nldas

will be updated, or that multilevel regression and post-stratification (described in Appendix F.2 below) can be used to handle small post-strata.

### 5.2.3 Collapsing post-strata

When a post-stratum is deemed too small, it is common to collapse/pool the post-stratum with other post-strata. Collapsing should be done among post-strata that are as similar as possible with respect to their distributions of field attributes. All decisions and rationale to collapse post-strata must be documented.

### 5.2.4 Alternative approaches for forming post-strata

As an alternative to collapsing post-strata, it is possible to form post-strata from quantiles of the response propensity, predicted mean, or a combination of the two (as described next) [22, 17, 23]. The response propensity can be estimated by regressing a response indicator[8] on the post-stratification variables, and the predicted mean can be estimated by regressing the emission reductions and removals[9] on the post-stratification variables. Post-strata can then be formed from quintiles (or other quantiles) of i) the response propensity, ii) the predicted mean, or iii) the cross-classification of quantiles of the response propensity and quantiles of the predicted mean. Forming post-strata from the response propensity controls nonresponse bias but not variance. In contrast, forming post-strata from the predicted mean controls both nonresponse bias and variance, but requires a separate model for each outcome variable (i.e., each type of emission reduction and removal) [17].

Logistic regression can be used to estimate response propensities, though other methods such as a random forest or gradient boosted trees can also be used. Linear regression can be used to estimate the predicted mean, though modeling assumptions should be verified with common modeling diagnostics [24]. If linear regression does not fit the data well as determined by model diagnostics, then generalized linear models may be explored as another modeling option.

If either response propensity and/or expected mean stratification is used, all decisions, modeling choices, and rationale must be documented.

While post-stratifying on the response propensity and/or predicted mean can reduce the number of post-strata, it can also make it difficult to discern which fields or sub-fields are grouped together. Therefore, if this approach is taken, descriptive summaries of the post-strata must be provided, including:

- Maps showing the boundaries of all post-strata
- A table showing marginal distribution of the same post-stratum attributes shown in Table 2

### 5.2.5 Potential changes in post-strata over time

This document considers the task of estimating a project's total emission reduction in one period of time. SEP projects report emission reductions on some cadence, such as annually. Project-wide emission reductions are estimated separately for each time period.

[8] When post-stratifying at the field level, the response indicator designates whether the entire field was missing. When post-stratifying at the sub-field level, the response indicator designates whether the point was missing

[9] When post-stratifying at the field level, the emission reductions/removals are the mean for the field. When post-stratifying at the sub-field level, the emission reductions/removals are for the individual point.

In estimates for subsequent periods of time, growers may adopt new practices that further enhance their sequestration and abatement of emissions, and weather may change substantially. To accommodate these changes, the set of post-stratification variables (Section 5.2.1) and the assignment of land to post-strata (Section 5.2.3 and 5.2.4) may change from one time period to the next.

### 5.2.6 Statistical estimators of totals and variances

This section covers the statistical estimators of the total GHG emission reduction and removal, as well as the variance, conditional on a selected post-stratification. The specific approach will depend on the project's sampling design and the level at which units are post-stratified (see Section 4). In this section, we provide a few examples for common sampling schemes, though others are possible.

**Example 3: Post-stratification at the field level in a two-stage sampling design where primary units (fields) are selected with PPSWR and secondary units (points) are selected by an SRSWR**

This is the same sampling design as in Appendix D.4 of the SEP [1]. In the first stage, $n^{\text{plan}}$ field selections are made with probability proportional to size with replacement (PPSWR), and in the second stage a simple random sample with replacement (SRSWR) of $m^{\text{plan}}$ points is drawn from each field selection. If field $i$ is selected $k_i$ times in the first stage, there would be $k_i \times m^{\text{plan}}$ selected points for soil samples from field $i$. If no data were missing, there would be $n^{\text{plan}} \times m^{\text{plan}}$ samples.

Suppose $H$ post-strata are formed at the field level with areas $A_h$, $h = 1, \ldots, H$. Let $n_h$ be the observed number of field selections in post-stratum $h$ with full or partial soil samples, and let $m_{hi}$ be the observed number of sampled points in field $i$ of post-stratum $h$. Note that due to nonresponse, the total observed number of soil samples may be less than originally planned, i.e., $\sum_{h=1}^{H} \sum_{i=1}^{n_h} m_{hi} \leq n^{\text{plan}} m^{\text{plan}}$. Let $y_{hij}$ be the GHG emission reductions/removals at point $j$ in field $i$ of post-stratum $h$, let $\bar{y}_{hi}$ be the average over the observed points in field $i$, and let $\bar{y}_h$ be the average over the observed field selections in post-stratum $h$. The post-stratified estimator of the total is

$$\hat{\tau}_{\text{ps}} = \sum_{h=1}^{H} A_h \bar{y}_h = \sum_{h=1}^{H} \left( \frac{A_h}{n_h} \sum_{i=1}^{n_h} \bar{y}_{hi} \right) = \sum_{h=1}^{H} \sum_{i=1}^{n_h} \sum_{j=1}^{m_{hi}} w_{hi} y_{hij} \tag{9}$$

where the weights are $w_{hi} = A_h/(n_h m_{hi})$.

To estimate the variance, it is typically recommended to condition on the sizes of the post-strata, which leads to the same calculation as when the strata are specified before sampling [13, 14, 25]:

$$\widehat{\text{Var}}(\hat{\tau}_{\text{ps}}) = \sum_{h=1}^{H} \frac{A_h^2}{n_h} s_h^2 \tag{10}$$

where

$$s_h^2 = \frac{1}{n_h - 1} \sum_{i=1}^{n_h} (\bar{y}_{hi} - \bar{y}_h)^2. \tag{11}$$

To estimate the variance of the areal mean $\hat{\mu}_{\text{ps}} = \hat{\tau}_{\text{ps}}/A$ where $A = \sum_{h=1}^{H} A_h$, all terms in (10) are divided by $A^2$.

**Example 4: Continuation of Example 3 when post-stratifying at the sub-field level**

This is a continuation of Example 3, though now suppose that post-strata are formed at the sub-field level. In this case, the estimate of variance is not straightforward because the primary sampling units are fields, but fields can be split across different post-strata. However, it is possible to obtain bootstrap or jackknife estimates of the variance [26, 27]. When conducting bootstrap or jackknife resampling, the units for resampling should be the primary units (fields) to mimic the original sample design, as these are the largest independent units (ultimate clusters per [4]).

For example, with bootstrap resampling, for each of $b = 1, \ldots, B$ resamples, a sample of $n$ of the observed field selections would be resampled with replacement to form the $b^{\text{th}}$ bootstrap resample. For each bootstrap resample, the within-strata means would be calculated, denoted as $\bar{y}_h^{(b)}$. Let

$$\hat{\tau}_{\text{ps}}^{(b)} = \sum_{h=1}^{H} A_h \bar{y}_h^{(b)} \tag{12}$$

be the $b^{\text{th}}$ bootstrap post-stratified estimate of the total, and let

$$\hat{\tau}_{\text{ps,boot}} = \frac{1}{B} \sum_{b=1}^{B} \hat{\tau}_{\text{ps}}^{(b)} \tag{13}$$

be the bootstrap estimate of the total. The bootstrap estimator of the variance of (13) is

$$v_{\text{boot}} = \frac{1}{1-B} \sum_{b=1}^{B} (\hat{\tau}_{\text{ps}}^{(b)} - \hat{\tau}_{\text{ps,boot}})^2 \tag{14}$$

The number of bootstrap resamples should be set at $B \geq 1,000$. Larger $B$ can improve the estimates [28], particularly for bootstrap confidence intervals, though $B = 1,000$ is typically sufficient for estimating bootstrap standard errors.

To estimate the variance of the areal mean $\hat{\mu}_{\text{ps,boot}} = \hat{\tau}_{\text{ps,boot}}/A$ where $A = \sum_{h=1}^{H} A_h$, all terms in (14) are divided by $A^2$.

As a consequence of bootstrap resampling at the field level, it is possible that in any given bootstrap resample there will be no observed soil samples in a given post-stratum, making it impossible to compute $\bar{y}_h^{(b)}$. If this occurs in only a few samples it may not be a concern, and the problematic bootstrap resample can be removed from the calculation of the variance (14). The number of resamples thrown out due to this problem should be reported for transparency, and it cannot exceed $30\% \times B$ per requirements in Section 5.2.2.

Example 5: Post-stratification of single-stage SRSWR

For single-stage SRSWR designs, post-stratification is straightforward at both the field and sub-field level because the points are the primary sampling units. An SRSWR can be drawn, for example, by sampling points at a constant density across the entire project area. Under this design, the number of points in a given field is a Poisson random variable for sufficiently large sample sizes. Consequently, this design is also called a Poisson sampling design with replacement.

Let $y_{hj}$ be the GHG emission reduction/removal at point $j$ in post-stratum $h$; let $n_h$ be the observed number of sampling points in post-stratum $h$; and let $\bar{y}_h = n_h^{-1} \sum_{j=1}^{n_h} y_{hj}$ be the mean in post-stratum $h$. Let $A_h$ be the area of post-stratum $h = 1, \ldots, H$. The post-stratified estimator of the total is

$$\hat{\tau}_{\text{ps}} = \sum_{h=1}^{H} A_h \bar{y}_h = \sum_{h=1}^{H} \left( \frac{A_h}{n_h} \sum_{j=1}^{n_h} y_{hj} \right) = \sum_{h=1}^{H} \sum_{j=1}^{n_h} w_h y_{hj} \tag{15}$$

where the weights are $w_h = A_h / n_h$. The post-stratified estimator of the variance is

$$\widehat{\text{Var}}(\hat{\tau}_{\text{ps}}) = \sum_{h=1}^{H} \frac{A_h^2}{n_h} s_h^2 \tag{16}$$

where

$$s_h^2 = \frac{1}{n_h - 1} \sum_{j=1}^{n_h} (y_{hj} - \bar{y}_h)^2. \tag{17}$$

To estimate the variance of the areal mean $\hat{\mu}_{\text{ps}} = \hat{\tau}_{\text{ps}} / A$ where $A = \sum_{h=1}^{H} A_h$, all terms in (16) are divided by $A^2$.

## 5.3 Imputation

For the purposes of modeling %SOC, biogeochemical model inputs may be imputed in the following scenarios:

- **%SOC**: If based on the guidance in Section 4.1, post-stratifying at the sub-field level is infeasible and post-stratifying at the field level is concerning because missingingess within fields may be related to emission reductions/removals, one may impute %SOC measurements in partially-sampled fields.
- **Bulk density, soil texture, and soil pH**: One may impute these soil attributes wherever they are missing. If %SOC is a strong predictor for the attribute but %SOC is not sampled at the given point, then imputed %SOC may be used as input to the imputation model so long as all imputations follow the guidance in Section 5.3.1 below.

Guidelines for when %SOC can be imputed are stricter than when bulk density, soil texture, and soil pH can be imputed for two reasons. First, %SOC tends to have more spatial variability

[5]. Second, growers are rewarded for increasing %SOC. (Although bulk density is important for calculating SOC stocks, an increase in bulk density does not mean carbon was sequestered [29].)

The remainder of this section provides guidance on how to impute soil measurements, and how to report performance metrics to provide transparency into the imputation results.

#### 5.3.1 Imputation approach and model selection

Imputation at a point may be done with a single draw from a predictive distribution [4, Ch 4.3]. Compared to conditional mean imputation, taking draws from the predictive distribution maintains similar within-field variability as would have been observed had there been no missing data (described below).

Any model type with a theoretical basis for taking draws from a predictive distribution can be used, including kriging, linear models, or generalized linear models, though other models may also be used. The predictive distribution can be based on a closed-form result, or it can use a resampling or simulation procedure. If imputing multiple soil attributes at the same point (e.g., %SOC and bulk density), one could consider using copulas to capture their correlation.

Project developers may use external datasets to train soil imputation models, including national soil surveys (e.g., SSURGO), remote sensing data (e.g., LiDAR for elevation), and soil samples collected outside of the project.

#### 5.3.2 Reporting imputation results and diagnostics

One must report summaries of 1) missing soil attributes, 2) variability of imputed values, 3) associations between soil attributes, and 4) performance of imputation models. The relevant summaries will depend on the sampling design.

The following summaries should be reported for a sampling design in which the plan is to measure %SOC at all randomly selected points, and to measure bulk density, soil texture, and soil pH at a random subset of selected points. The relevant summaries may be different for other sampling designs.

1. **Summaries of the number of missing measurements**: for each soil attribute that was imputed, report:
   (a) The number of partially sampled fields (i.e., fields with at least one point at which that attribute was imputed)
   (b) Histogram and summary statistics (including mean, median, min, max, and 10th, 25th, 75th, and 90th percentiles) of the distribution (across fields) of the proportion of points within the field at which the attribute is imputed
2. **Variability of imputed values**:
   (a) For %SOC: within-field variability of %SOC (i.e., within-field sample variance, averaged across fields) (i) fields with at least 1 imputed %SOC value, and (ii) in fully-sampled fields. The variability of (i) should be at least as large as the variability of (ii).
   (b) For bulk density, soil texture, and soil pH: Global variability of (i) imputed values and (ii) measured values. The variability of (i) should be at least as large as the variability of (ii).

3. **Associations between soil attributes**:

   (a) Spearman's correlation of %SOC and bulk density for (i) points where both are measured, (ii) points where either or both of %SOC and bulk density are imputed. The two correlations should be similar.

4. **Performance of imputation model**:

   (a) For all soil attributes: Performance of the imputation model's predictive draws on held-out data (bias, root mean squared error) in cross validation experiments, so that verifiers can see how close absolute bias is to zero and how large RMSE is.

# 6 Statistical adjustment methods using predictions from Bayesian-calibrated models

This section builds on Section 5 by providing details on how to conduct statistical adjustments when using predictions from Bayesian-calibrated models. As noted in Section 5, the estimators in Examples 1 through 5 are applicable when GHG emission reductions/removals are predicted with biogeochemical models for which calibration is not necessary, but in practice biogeochemical models typically need to go through a calibration process before being used in an SEP project.[10] While necessary for minimizing bias, the calibration process can introduce additional uncertainty that must be accounted for.

Section 6.1 gives a brief overview of Bayesian calibration, and Section 6.2 provides details on how to extend the examples in Sections 5.1 and 5.2.6 to SEP projects using Bayesian-calibrated biogeochemical models. Section 6 focuses on the statistical aspects of using Bayesian-calibrated models to estimate GHG emission reductions/removals, because all other aspects are covered in Section 5; all guidance and requirements in Section 5 (other than the example calculations in Sections 5.1 and 5.2.6) must also be followed when using the methods described in Section 6.

## 6.1 Overview of Bayesian calibration of biogeochemical models

Biogeochemical models such as DayCent [30] are typically deterministic functions of covariates (e.g., initial soil organic carbon concentration (%SOC), soil texture, pH, and bulk density) and parameters (e.g., maximum decomposition rates for carbon pools, and the effect of moisture and tillage on decomposition rates). There are typically many parameters that are theoretically constant, though their values are unknown.

In general, model calibration (distinct from covariate calibration described in Appendix F.1) is the process of using calibration data (e.g. %SOC as measured in an experimental study, denoted as $z$) to select parameter values that result in accurate biogeochemical model predictions. However, model calibration is an estimation process, so there is inherently uncertainty in whether the correct parameter values have been selected. Bayesian model calibration [31] quantifies this uncertainty by selecting a distribution of parameter values (the *posterior* distribution from a Bayesian analysis) instead of a single value. As described in Appendix C, instead of a single prediction of GHG emission reductions/removals, Bayesian-calibrated models generate $L$ Monte Carlo (MC) draws from

[10] See the *Requirements and Guidance for Model Calibration, Validation, Uncertainty, and Verification* appendix to the SEP [1]

a *posterior predictive distribution* of GHG emission reductions/removals. Following the notation in Table 6 in Appendix B, for one-stage sampling designs these MC draws are denoted as $\tilde{y}_{j\ell}$ for point $j$ and MC draw $\ell$, which are *realizations* of the *predictive random variable* $\tilde{Y}_j$. Please see Appendices B and C for details.

Let $\tau$ be the true population total GHG emission reductions/removals in the form of $tCO_2e$ of $CO_2$, $N_2O$, and $CH_4$ molecules that are not in the atmosphere due to the land management practice changes of growers participating in the SEP project. The sources of uncertainty in estimating $\tau$ with Bayesian-calibrated models include:

1. Potential biogeochemical model prediction errors:
    - Potential calibration errors, i.e., the posterior distribution is not concentrated on the truth
    - Potential model inadequacy (see Kennedy and O'Hagan [31])
2. Potential measurement error in calibration data (both model inputs and measured outcomes)
3. Potential measurement error in model inputs used to make predictions outside of the calibration data
4. Sampling variability due to making model predictions at a random selection of fields/points in the SEP project

The examples in the remainder of this section describe how to estimate the variance in a way that accounts for all the sources of uncertainty noted above for a few common sampling designs. The first two sources of uncertainty (biogeochemical model prediction errors and calibration data measurement error) are accounted for by taking draws from a posterior predictive distribution that incorporates both sources of variability, and are captured by the $s^2_{\text{model}}$ term in the examples below. The third and fourth source of uncertainty (model input measurement errors and sampling variability) are accounted for through the sample variance and is captured by the $s^2_{\text{sample}}$ term in the examples below.[11] For details and derivations, please see Appendices C and D, and for more details on notation please see Appendix B.

## 6.2 Estimation of total and variance using Bayesian-calibrated model predictions

### 6.2.1 Unadjusted analyses

As described in section 2, an unadjusted analysis is always required for comparison to adjusted analyses. An unadjusted analysis may also be provided as the sole analysis if there is no concern over potential nonresponse bias as discussed in Section 3.2.

In an unadjusted analysis, the observed data are analyzed as though there were no missing data. The exact calculations will depend on the sampling design, though we provide examples below.

[11] Similar to Appendix D of the SEP [1], we assume that measurement errors in model inputs translate to uncorrelated measurement errors in model predictions, which is accounted for through the sampling variance [12, Eq. 13.43]

Example 6: Unadjusted analysis of a one-stage sampling design with Bayesian-calibrated model predictions where points are selected with a simple random sample with replacement (SRSWR)

Per Tables 6 and 8 in Appendix B, let $\tilde{Y}_j$ be the predictive random variable for the GHG emission reductions/removals for observed points $j = 1, \ldots, n$. Also, let $\tilde{y}_{j\ell}$, $\ell = 1, \ldots, L$ be MC draws from the posterior predictive distribution of $\tilde{Y}_j$ conditional on the calibration data $\boldsymbol{z} = (z_1, \ldots, z_{n_{\text{cal}}})$, and let $A$ be the total area of the SEP project. All $j = 1, \ldots, n$ sampled points are outside of the calibration data set.
We can construct a posterior predictive random variable of the population total as

$$\tilde{\tau} = \frac{A}{n} \sum_{j=1}^{n} \tilde{Y}_j. \tag{18}$$

As described in Appendix D, we can estimate the total conditional on the calibration data as

$$\hat{\tau} = \hat{\mathbb{E}}[\tilde{\tau}|\boldsymbol{z}] = \frac{A}{n} \sum_{j=1}^{n} \bar{\tilde{y}}_j \tag{19}$$

where $\bar{\tilde{y}}_j = L^{-1} \sum_{\ell=1}^{L} \tilde{y}_{j\ell}$. Let

$$\tilde{\tau}_\ell = \frac{A}{n} \sum_{j=1}^{n} y_{j\ell} \tag{20}$$

be the $\ell^{th}$ draw from the posterior predictive distribution of the population total. The estimator of variance is

$$\widehat{\text{Var}}(\tilde{\tau}|\boldsymbol{z}) = \frac{A^2}{n} s^2_{\text{sample}} + s^2_{\text{model}} \tag{21}$$

where

$$s^2_{\text{sample}} = \frac{1}{n-1} \sum_{j=1}^{n} (\bar{\tilde{y}}_j - \hat{\mu})^2 \tag{22a}$$

$$s^2_{\text{model}} = \frac{1}{L-1} \sum_{\ell=1}^{L} (\tilde{\tau}_\ell - \hat{\tau})^2 \tag{22b}$$

and $\hat{\mu} = \hat{\tau}/A$ and $\hat{\tau} = L^{-1} \sum_{\ell=1}^{L} \tilde{\tau}_\ell = (A/n) \sum_{j=1}^{n} \bar{\tilde{y}}_j$. To estimate the variance for the areal mean $\hat{\mu}$, both terms in (21) are divided by $A^2$.

Example 7: Unadjusted analysis of a two-stage sampling design with Bayesian-calibrated model predictions where fields are selected with probability proportional to size (PPSWR) and points are selected with a simple random sample with replacement (SRSWR)

This is the same as Example 1 but where outcomes are predicted with a Bayesian-calibrated model. Per Tables 7 and 8 in Appendix B, let $\tilde{Y}_{ij}$ be the predictive random variable for the GHG emission reductions/removals for field selection $i = 1, \ldots, n$ and observed point $j = 1, \ldots, m_i$, and let $\tilde{y}_{ij\ell}$, $\ell = 1, \ldots, L$ be MC draws from the posterior predictive distribution of $\tilde{Y}_{ij}$ conditional on the calibration data $\boldsymbol{z} = (z_1, \ldots, z_{n_{\text{cal}}})$. All $i = 1, \ldots, n$ sampled fields are outside of the calibration data set.

The probability of selecting field $i$ in each draw is $p_i = A_i/A$ where $A = \sum_{i=1}^{N} A_i$. We can construct a Hansen-Hurwitz type posterior predictive random variable of the population total as

$$\tilde{\tau} = \frac{1}{n} \sum_{i=1}^{n} \frac{A_i}{p_i} \bar{\tilde{Y}}_i = \frac{A}{n} \sum_{i=1}^{n} \bar{\tilde{Y}}_i \tag{23}$$

where $\bar{\tilde{Y}}_i = m_i^{-1} \sum_{j=1}^{m_i} \tilde{Y}_{ij}$.

As described in Appendix D, we can estimate the total conditional on the calibration data as

$$\hat{\tau} = \hat{\mathbb{E}}[\tilde{\tau}|\boldsymbol{z}] = \frac{A}{n} \sum_{i=1}^{n} \bar{\tilde{y}}_i \tag{24}$$

where $\bar{\tilde{y}}_i = m_i^{-1} \sum_{j=1}^{m_i} \bar{\tilde{y}}_{ij}$, and $\bar{\tilde{y}}_{ij} = L^{-1} \sum_{\ell=1}^{L} \tilde{y}_{ij\ell}$. Let

$$\tilde{\tau}_\ell = \frac{A}{n} \sum_{i=1}^{n} \left( \frac{1}{m_i} \sum_{j=1}^{m_i} y_{ij\ell} \right) \tag{25}$$

be the $\ell^{th}$ draw from the posterior predictive distribution of the population total. The estimator of variance is

$$\widehat{\text{Var}}(\tilde{\tau}|\boldsymbol{z}) = \frac{A^2}{n} s^2_{\text{sample}} + s^2_{\text{model}} \tag{26}$$

where

$$s^2_{\text{sample}} = \frac{1}{n-1} \sum_{i=1}^{n} (\bar{\tilde{y}}_i - \hat{\mu})^2 \tag{27a}$$

$$s^2_{\text{model}} = \frac{1}{L-1} \sum_{\ell=1}^{L} (\tilde{\tau}_\ell - \hat{\tau})^2 \tag{27b}$$

and $\hat{\mu} = \hat{\tau}/A$ and $\hat{\tau} = L^{-1} \sum_{\ell=1}^{L} \tilde{\tau}_\ell = (A/n) \sum_{i=1}^{n} \bar{\tilde{y}}_i$. To obtain the variance of the areal mean $\hat{\mu}$, both terms in (26) are divided by $A^2$.

Example 8: Continuation of example 7 but where fields are selected with probability proportional to estimated size

This is the same as Example 7 but where fields in the first stage are selected with probability proportional to estimated size $\hat{A}_i$, $i = 1 \ldots, N$. The probability of selecting field $i$ in each draw is $p_i = \hat{A}_i/\hat{A}$ where $\hat{A} = \sum_{i=1}^{N} \hat{A}_i$. Similar to Example 2, we construct a Hansen-Hurwitz type posterior predictive random variable of the population total as

$$\tilde{\tau} = \frac{1}{n} \sum_{i=1}^{n} \frac{A_i}{p_i} \bar{\tilde{Y}}_i = \frac{\hat{A}}{n} \sum_{i=1}^{n} \frac{A_i}{\hat{A}_i} \bar{\tilde{Y}}_i \tag{28}$$

where $\bar{\tilde{Y}}_i = m_i^{-1} \sum_{j=1}^{m_i} \tilde{Y}_{ij}$.
As described in Appendix D, we can estimate the total conditional on the calibration data $\boldsymbol{z}$ as

$$\hat{\tau} = \hat{\mathbb{E}}[\tilde{\tau}|\boldsymbol{z}] = \frac{\hat{A}}{n} \sum_{i=1}^{n} \frac{A_i}{\hat{A}_i} \bar{\tilde{y}}_i \tag{29}$$

where $\bar{\tilde{y}}_i = m_i^{-1} \sum_{j=1}^{m_i} \bar{\tilde{y}}_{ij}$, and $\bar{\tilde{y}}_{ij} = L^{-1} \sum_{\ell=1}^{L} \tilde{y}_{ij\ell}$. Let

$$\tilde{\tau}_\ell = \frac{\hat{A}}{n} \sum_{i=1}^{n} \left( \frac{A_i}{\hat{A}_i m_i} \sum_{j=1}^{m_i} y_{ij\ell} \right) \tag{30}$$

be the $\ell^{th}$ draw from the posterior predictive distribution of the population total. The estimator of variance is

$$\widehat{\mathrm{Var}}(\tilde{\tau}|\boldsymbol{z}) = \frac{\hat{A}^2}{n} s^2_{\text{sample}} + s^2_{\text{model}} \tag{31}$$

where

$$s^2_{\text{sample}} = \frac{1}{n-1} \sum_{i=1}^{n} \left( \frac{A_i}{\hat{A}_i} \bar{\tilde{y}}_i - \hat{\mu} \right)^2 \tag{32a}$$

$$s^2_{\text{model}} = \frac{1}{L-1} \sum_{\ell=1}^{L} (\tilde{\tau}_\ell - \hat{\tau})^2 \tag{32b}$$

and $\hat{\mu} = \hat{\tau}/A$ and $\hat{\tau} = L^{-1} \sum_{\ell=1}^{L} \tilde{\tau}_\ell = (\hat{A}/n) \sum_{i=1}^{n} (A_i/\hat{A}_i) \bar{\tilde{y}}_i$. To estimate the variance of the areal mean $\hat{\mu}$, both terms in (31) are divided by $A^2$.

### 6.2.2 Adjusted analyses

This section covers the statistical estimators of the total GHG emission reduction and removal, as well as the variance, conditional on a selected post-stratification. The specific approach will depend on the project's sampling design and the level at which units are post-stratified (see Section 4). In

this section, we provide a few examples for common sampling schemes, though others are possible.

**Example 9: Post-stratification at the field level with Bayesian-calibrated predictions in a two-stage sampling design where primary units (fields) are selected with PPSWR and secondary units (points) are selected by a SRSWR**

This is the same as Example 3 but where outcomes are predicted with a Bayesian-calibrated model. Per Tables 7 and 8 in Appendix B, let $\tilde{Y}_{hij}$ be the predictive random variable of GHG emission reductions/removals at observed point $j = 1, \ldots, m_i$ in field selection $i = 1, \ldots, n_h$ of post-stratum $h = 1, \ldots, H$, and let $\tilde{y}_{hij\ell}$, $\ell = 1, \ldots, L$, be an MC draw from the posterior predictive distribution of $\tilde{Y}_{hij}$.
Applying the results of Example 6 within each stratum, where fields are the primary sampling units instead of points, we can construct a posterior predictive random variable for the population total in post-stratum $h$ as

$$\tilde{\tau}_h = \frac{A_h}{n_h} \sum_{i=1}^{n_h} \bar{\tilde{Y}}_{hi} \tag{33}$$

where $\bar{\tilde{Y}}_{hi} = m_i^{-1} \sum_{j=1}^{m_i} \tilde{Y}_{hij}$. Also, let $\tilde{\tau}_{\text{ps}} = \sum_{h=1}^{H} \tilde{\tau}_h$ be the predictive random variable of the post-stratified population total.
We can estimate the population total in stratum $h$ conditional on the calibration data as

$$\hat{\tau}_h = \hat{\mathbb{E}}\left[\tilde{\tau}_h | \boldsymbol{z}\right] = \frac{A_h}{n_h} \sum_{i=1}^{n_h} \bar{\tilde{y}}_{hi} \tag{34}$$

where $\bar{\tilde{y}}_{hi} = m_i^{-1} \sum_{j=1}^{m_i} \bar{\tilde{y}}_{hij}$ and $\bar{\tilde{y}}_{hij} = L^{-1} \sum_{\ell=1}^{L} \tilde{y}_{hij\ell}$. The post-stratified estimator of the total is

$$\hat{\tau}_{\text{ps}} = \sum_{h=1}^{H} \hat{\tau}_h = \sum_{h=1}^{H} \left( \frac{A_h}{n_h} \sum_{i=1}^{n_h} \bar{\tilde{y}}_{hi} \right). \tag{35}$$

Let

$$\tilde{\tau}_{h\ell} = \frac{A_h}{n_h} \sum_{i=1}^{n_h} \left( \frac{1}{m_i} \sum_{j=1}^{m_i} \tilde{y}_{hij\ell} \right) \tag{36}$$

be the $\ell^{th}$ MC draw of the population total in post-straum $h$, and let

$$\tilde{\tau}_{\text{ps},\ell} = \sum_{h=1}^{H} \tilde{\tau}_{h\ell}. \tag{37}$$

To estimate the variance, it is typically recommended to condition on the size of the post-strata [13, 14, 25], which gives

$$\widehat{\text{Var}}(\tilde{\tau}_{\text{ps}} | \boldsymbol{z}) = \sum_{h=1}^{H} \left\{ \frac{A_h^2}{n_h} s^2_{\text{sample},h} \right\} + s^2_{\text{model}} \tag{38}$$

where

$$s^2_{\text{sample},h} = \frac{1}{n_h - 1}\sum_{i=1}^{n_h}(\bar{\tilde{y}}_{hi} - \hat{\mu}_h)^2 \tag{39a}$$

$$s^2_{\text{model}} = \frac{1}{L-1}\sum_{\ell=1}^{L}(\tilde{\tau}_{\text{ps},\ell} - \hat{\tau}_{\text{ps}})^2 \tag{39b}$$

and $\hat{\mu}_h = \hat{\tau}_h / A_h$.
The calculation of model variance $s^2_{\text{model}}$ cannot be split across strata because the same MC draws can be used across strata, which can induce dependence; while the field selections are independent between strata, the MC draws may not be.
To estimate the variance of the areal mean $\hat{\mu}_{\text{ps}} = \hat{\tau}_{\text{ps}}/A$, all terms in (38) are divided by $A^2$.

**Example 10: Continuation of Example 9 when post-stratifying at the sub-field level**

This is a continuation of Example 9, though now suppose that post-strata are formed at the sub-field level. Because primary sampling units (fields) can be split across multiple post-strata, we drop the subscript $i$ for fields and re-index observed points in stratum $h$ as $j = 1, \ldots, n_h$.
As in Example 4, the estimate of variance is not straightforward because the primary sampling units can be split across different post-strata. In particular, as described in Appendix D, we can no longer just estimate $s^2_{\text{sample}}$ within each post-stratum. However, as also described in Appendix D we can estimate variability through a bootstrap or jackknife procedure [26, 27]. When conducting bootstrap or jackknife resampling, the units for resampling should be the primary units (fields) to mimic the original sample design, as these are the largest independent units (ultimate clusters per [4]).
For example, with bootstrap resampling, for each of $b = 1, \ldots, B$ resamples, a sample of $n$ of the observed field selections would be resampled with replacement to form the $b^{\text{th}}$ bootstrap resample. For each bootstrap resample, let

$$\tilde{\tau}^{(b)}_{\text{ps},\ell} = \sum_{h=1}^{H}\frac{A_h}{n_h^{(b)}}\sum_{j=1}^{n_h^{(b)}}\tilde{y}^{(b)}_{hj\ell} \tag{40}$$

be the $\ell^{th}$ draw from the posterior predictive distribution of the population total. The $b^{\text{th}}$ bootstrap post-stratified estimate of the total is

$$\hat{\tau}^{(b)}_{\text{ps}} = \frac{1}{L}\sum_{\ell=1}^{L}\tilde{\tau}^{(b)}_{\text{ps},\ell} \tag{41a}$$

$$= \sum_{h=1}^{H}\frac{A_h}{n_h^{(b)}}\sum_{j=1}^{n_h^{(b)}}\bar{\tilde{y}}^{(b)}_{hj} \tag{41b}$$

where $\bar{\tilde{y}}^{(b)}_{hj} = L^{-1}\sum_{\ell=1}^{L}\tilde{y}^{(b)}_{hj\ell}$.

The variance estimator for the $b^{th}$ bootstrap resample is

$$v^{(b)} = \frac{1}{L-1}\sum_{\ell=1}^{L}\left(\tilde{\tau}_{\text{ps},\ell}^{(b)} - \hat{\tau}_{\text{ps}}^{(b)}\right)^2. \tag{42}$$

Let

$$\hat{\tau}_{\text{ps,boot}} = \frac{1}{B}\sum_{b=1}^{B}\hat{\tau}_{\text{ps}}^{(b)} \tag{43}$$

be the mean of the bootstrap totals.
As shown in Appendix D, the variance of $\hat{\tau}_{\text{ps,boot}}$ can be estimated as

$$\widehat{\text{Var}}(\hat{\tau}_{\text{ps,boot}}|\boldsymbol{z}) = v_{\text{sample}} + v_{\text{model}} \tag{44}$$

where

$$v_{\text{sample}} = \frac{1}{B-1}\sum_{b=1}^{B}(\hat{\tau}_{\text{ps}}^{(b)} - \hat{\tau}_{\text{ps,boot}})^2 \tag{45a}$$

$$v_{\text{model}} = \frac{1}{B}\sum_{b=1}^{B}v^{(b)}. \tag{45b}$$

To estimate the variance of the areal mean $\hat{\mu}_{\text{ps,boot}} = \hat{\tau}_{\text{ps,boot}}/A$, all terms in (44) are divided by $A^2$.
The number of bootstrap resamples should be set at $B \geq 1,000$. Larger $B$ can improve the estimates [28], particularly for bootstrap confidence intervals, though $B = 1,000$ is typically sufficient for estimating bootstrap standard errors.
As a consequence of resampling at the field level, it is possible that in any given bootstrap resample there will be no observed soil samples in a given post-stratum, making it impossible to compute the total and variance for that stratum. If this occurs in only a few bootstrap samples it may not be a concern, and the problematic bootstrap resample can be removed from the calculation of the variance (44) (the number $B$ in the denominator should be equal to the number of bootstrap samples that were not thrown out). The number of resamples thrown out due to this problem should be reported for transparency, and it cannot exceed $30\% \times B$ per requirements in Section 5.2.2.

**Example 11: Post-stratification of single-stage SRSWR with Bayesian-calibrated model predictions**

This example is the same as Example 5 but with Bayesian-calibrated model predictions. Per Tables 6 and 8 in Appendix B, let $\tilde{Y}_{hj}$ be the predictive random variable for the GHG emission reductions/removals for observed point $j = 1, \ldots, n_h$ in post-stratum $h = 1, \ldots, H$. Also, let $\tilde{y}_{hj\ell}$, $\ell = 1, \ldots, L$ be MC draws from the posterior predictive distribution of $\tilde{Y}_{hj}$ conditional on the calibration data $\boldsymbol{z} = (z_1, \ldots, z_{n_{\text{cal}}})$, and let $A_h$ be the total area of post-stratum $h$. All $j = 1, \ldots, n$ sampled points are outside of the calibration data set.

We can construct a posterior predictive random variable of the total in stratum $h$ as

$$\tilde{\tau}_h = \frac{A_h}{n_h} \sum_{j=1}^{n} \tilde{Y}_{hj}. \tag{46}$$

Also, let $\tilde{\tau}_{\text{ps}} = \sum_{h=1}^{H} \tilde{\tau}_h$ be the post-stratified predictive random variable of the population total. As described in Appendix D, we can estimate the total conditional on the calibration data as

$$\hat{\tau}_h = \hat{\mathbb{E}}[\tilde{\tau}_h | \boldsymbol{z}] = \frac{A_h}{n_h} \sum_{j=1}^{n_h} \bar{\tilde{y}}_{hj} \tag{47}$$

where $\bar{\tilde{y}}_{hj} = L^{-1} \sum_{\ell=1}^{L} \tilde{y}_{hj\ell}$. The post-stratified estimator of the population total is

$$\hat{\tau}_{\text{ps}} = \sum_{h=1}^{H} \hat{\tau}_h = \sum_{h=1}^{H} \left( \frac{A_h}{n_h} \sum_{j=1}^{n_h} \bar{\tilde{y}}_{hj} \right). \tag{48}$$

Let

$$\tilde{\tau}_{h\ell} = \frac{A_h}{n_h} \sum_{j=1}^{n_h} y_{hj\ell} \tag{49}$$

be the $\ell^{th}$ draw from the posterior predictive distribution of the population total in post-stratum $h$, and let

$$\tilde{\tau}_{\text{ps},\ell} = \sum_{h=1}^{H} \tilde{\tau}_{h\ell}. \tag{50}$$

To estimate the variance, it is typically recommended to condition on the size of the post-strata [13, 14, 25], which gives

$$\widehat{\text{Var}}(\tilde{\tau}_{\text{ps}} | \boldsymbol{z}) = \sum_{h=1}^{H} \left\{ \frac{A_h^2}{n_h} s^2_{\text{sample},h} \right\} + s^2_{\text{model}} \tag{51}$$

where

$$s^2_{\text{sample},h} = \frac{1}{n_h - 1} \sum_{j=1}^{n_h} (\bar{\tilde{y}}_{hj} - \hat{\mu}_h)^2 \tag{52a}$$

$$s^2_{\text{model}} = \frac{1}{L - 1} \sum_{\ell=1}^{L} (\tilde{\tau}_{\text{ps},\ell} - \hat{\tau}_{\text{ps}})^2 \tag{52b}$$

and $\hat{\mu}_h = \hat{\tau}_h / A_h$.
As in Example 9, the calculation of model variance $s^2_{\text{model}}$ cannot be split across strata because the same MC draws can be used across strata, which can induce dependence; while the field selections are independent between strata, the MC draws may not be.
To estimate the variance of the areal mean $\hat{\mu}_{\text{ps}} = \hat{\tau}_{\text{ps}} / A$, all terms in (51) are divided by $A^2$.

### 6.2.3 Combining estimates across pre-strata

SEP projects may *pre-stratify* the population [1, Table 6.2], meaning that they partition the population into non-overlapping subsets and select sample points within each pre-stratum, independently of the other pre-strata. The sample design may vary from one pre-stratum to another, and there may be post-strata within each pre-stratum. Pre-stratification can reduce uncertainty without increasing the sample size by allocating more sampling to more heterogeneonus pre-strata. The following example shows how to combine the estimates from the pre-strata.

**Example 12: Pre-stratification**

The project's spatial boundary is partitioned into $G$ non-overlapping pre-strata, indexed by $g = 1, \ldots, G$. Within each pre-stratum, sample points are selected independently of the other pre-strata. Let $A_g$ be the area of pre-stratum $g$. (In previous examples, we suppressed the index $g$ for pre-strata because all calculations were specific to a single pre-stratum. If post-stratification is done within pre-stratum $g$, then the total area in pre-stratum $g$ is equal to the sum of the post-strata areas within pre-stratum $g$, i.e. $A_g = \sum_h A_{gh}$ where $A_{gh}$ is the area of post-stratum $h$ within pre-stratum $g$.) Because pre-strata do not spatially overlap, the total area of the project, $A$, equals the sum $\sum_{g=1}^{G} A_g$.
We can construct a posterior predictive random variable of the population total as the sum of the total in each pre-stratum:

$$\tilde{\tau} = \sum_{g=1}^{G} \tilde{\tau}_g, \tag{53}$$

where $\tilde{\tau}_g$ is the posterior predictive random variable of the population total in pre-stratum $g$, constructed using the example above that is appropriate for the sample design deployed in pre-stratum $g$. Similar to the area $A_g$, if post-stratification is used within pre-stratum $g$, then $\tilde{\tau}_g = \sum_h \tilde{\tau}_{hg}$ where $\tilde{\tau}_{hg}$ is the total within post-stratum $h$ of pre-stratum $g$. We can estimate the total conditional on the calibration data as the sum

$$\hat{\tau} = \hat{\mathbb{E}}\left[\tilde{\tau}|\boldsymbol{z}\right] = \sum_{g=1}^{G} \hat{\tau}_g, \tag{54}$$

where $\hat{\tau}_g$ is the estimate appropriate for the sample design in pre-stratum $g$ (summed over any post-strata within pre-stratum $g$, if applicable).
Let $\tilde{\tau}_{g\ell}$ be the $\ell^{\text{th}}$ draw from the posterior predictive distribution of the total in pre-stratum $g$, and let $\tilde{\tau}_\ell = \sum_{g=1}^{G} \tilde{\tau}_{g\ell}$ be the $\ell^{\text{th}}$ draw from the posterior predictive distribution of the population total.
Let $v_{\text{sample},g}$ be the estimated sampling variance for stratum $g$. $v_{\text{sample},g}$ can be retrieved from the example above that was deployed in pre-stratum $g$; for example, for the unadjusted SRSWR (Example 6), $v_{\text{sample},g}$ is the first term on the right-hand of Eq. (21) with the pre-stratum label $g$ attached to all the variables: $\left(A_g^2/n_g\right) s^2_{\text{sample},g}$.
The estimator of variance is

$$\widehat{\text{Var}}(\tilde{\tau}|\boldsymbol{z}) = v_{\text{sample}} + s^2_{\text{model}}, \tag{55}$$

where

$$v_{\text{sample}} = \sum_{g=1}^{G} v_{\text{sample},g} \tag{56a}$$

$$v_{\text{model}} = \frac{1}{L-1} \sum_{\ell=1}^{L} (\tilde{\tau}_\ell - \hat{\tau})^2 \tag{56b}$$

and the estimated areal mean is $\hat{\mu} = \hat{\tau}/A$. To estimate the variance for the areal mean $\hat{\mu}$, both terms in (55) are divided by $A^2$.

In Eq. (56a), the sampling variance of the total is the sum of the sampling variances in the individual pre-strata. The reason is that the sample locations are chosen independently across pre-strata (see, e.g., [12, Theorem 5.2]).

# Appendices

## A Multivariate standardized difference

This appendix provides details on how to compute multivariate standardized differences for categorical variables with $L > 2$ categories. As in Section 3.2, let $p_{\text{all},k\ell}$ and $p_{\text{obs},k\ell}$ be the proportion of points in category $\ell$ for attribute $k$ among i) all points in the planned sample, and ii) points with observed %SOC, respectively. Let $\boldsymbol{p}_{\text{all},k} = (p_{\text{all},k2}, \ldots, p_{\text{all},kL})$ and $\boldsymbol{p}_{\text{obs},k} = (p_{\text{obs},k2}, \ldots, p_{\text{obs},kL})$. The first category is dropped because it is fully determined by the remaining $L-1$ categories ($\sum_{\ell=1}^{L} p_{\text{all},k\ell} = 1$ and $\sum_{\ell=1}^{L} p_{\text{obs},k\ell} = 1$).[12] Also let $V_k$ be an $(L-1) \times (L-1)$ matrix of variance-covariances where the element in row $\ell$ and column $\ell'$ of $V_k$ is

$$V_{K\ell\ell'} = \begin{cases} p_{\text{all},k\ell}(1 - p_{\text{all},k\ell}) & \text{if } \ell = \ell' \\ -p_{\text{all},k\ell}\, p_{\text{all},k\ell'} & \text{if } \ell \neq \ell' \end{cases} \tag{57}$$

for $\ell, \ell' = 2, \ldots, L$. The multivariate standardized difference for attribute $k$ is defined as [35]

$$d_k = \sqrt{(\boldsymbol{p}_{\text{obs},k} - \boldsymbol{p}_{\text{all},k})^\intercal V_k^{-1} (\boldsymbol{p}_{\text{obs},k} - \boldsymbol{p}_{\text{all},k})}. \tag{58}$$

For $L = 2$ categories, (58) reduces to the absolute value of the univariate standardized difference given by (1).

## B Notation for Bayesian-calibrated models

Here, we summarize the notation used in Section 6 and Appendices C, D and E, and discuss the rationale for some choices.

[12] This is also the convention followed by Yang and Dalton [32] and the `tableone` packages for Python [33] and R [34].

## B.1 Calibration data

Table 5: Notation for calibration data

| Quantity | Notation | Units |
|---|---|---|
| Measured GHG emissions | $z_i$ | $tCO_2e$/acre |
| Covariates | $\boldsymbol{x}_i$ | |
| Subscript for calibration sites/treatment | $i = 1, \ldots, n_{\text{cal}}$ | |

## B.2 Posterior predictions

In Bayesian literature, outcomes that are not yet observed, such as GHG emissions outside of the calibration data, are typically denoted with a tilde decorator such as $\tilde{Z}$ for random variables and $\tilde{z}$ for realizations [36, 37]. Observed covariates associated with unobserved outcomes (e.g., initial %SOC, soil texture, pH, and bulk density at sampled locations) are also commonly decorated with a tilde such as $\tilde{\boldsymbol{x}}$ [37]. We follow the same conventions.

The predictive random variable $\tilde{Z}$ follows a *posterior predictive distribution* that [36]:

1. Does not depend on any unknown quantities, such as biogeochemical model parameters (denoted as $\boldsymbol{\theta}$ in Table 8), and

2. Does depend on the observed data (calibration data described in Table 5).

Bayesian-calibrated biogeochemical models make extensive use of posterior predictions to estimate GHG emission reductions/removals.

Per the SEP [1], the baseline is an average of several different crop patterns, and the predictive random variable $\tilde{Z}_{\text{bl}}$ in Tables 6 and 7 represents an average of these baseline scenarios.

Table 6: Notation specific to one-stage sampling designs

| Quantity | Notation | Units |
|---|---|---|
| Predictive random variables | | |
| GHG emissions for project, baseline | $\tilde{Z}_{\text{pr},j}$, $\tilde{Z}_{\text{bl},j}$ | $tCO_2e$/acre |
| GHG emission reductions | $\tilde{Y}_j = \tilde{Z}_{\text{pr},j} - \tilde{Z}_{\text{bl},j}$ | $tCO_2e$/acre |
| Draws from posterior predictive distribution | | |
| GHG emissions for project, baseline | $\tilde{z}_{\text{pr},j\ell}$, $\tilde{z}_{\text{bl},j\ell}$ | $tCO_2e$/acre |
| GHG emission reductions | $\tilde{y}_{j\ell} = \tilde{z}_{\text{pr},j\ell} - \tilde{z}_{\text{bl},j\ell}$ | $tCO_2e$/acre |
| Total area | $A$ | acres |
| Covariates for project, baseline | $\tilde{\boldsymbol{x}}_{\text{pr},j}$, $\tilde{\boldsymbol{x}}_{\text{bl},j}$ | |
| Total number of points in population | Not applicable | |
| Observed sample vector | $\boldsymbol{s}$ | |
| Random sample vector | $\boldsymbol{S}$ | |
| Subscript for sampled points | $j = 1, \ldots, n$ | |
| Subscript for MC draw | $\ell = 1, \ldots, L$ | |

Table 7: Notation specific to two-stage sampling designs

| Quantity | Notation | Units |
|---|---|---|
| Predictive random variables | | |
| GHG emissions for project, baseline | $\tilde{Z}_{\text{pr},ij}$, $\tilde{Z}_{\text{bl},ij}$ | $tCO_2e$/acre |
| GHG emission reductions | $\tilde{Y}_{ij} = \tilde{Z}_{\text{pr},ij} - \tilde{Z}_{\text{bl},ij}$ | $tCO_2e$/acre |
| Draws from posterior predictive distribution | | |
| GHG emissions for project, baseline | $\tilde{z}_{\text{pr},ij\ell}$, $\tilde{z}_{\text{bl},ij\ell}$ | $tCO_2e$/acre |
| GHG emission reductions | $\tilde{y}_{ij\ell} = \tilde{z}_{\text{pr},ij\ell} - \tilde{z}_{\text{bl},ij\ell}$ | $tCO_2e$/acre |
| Area of field | $A_i$ | acres |
| Estimated/approximate area of field | $\hat{A}_i$ | acres |
| Covariates for project, baseline | $\tilde{\boldsymbol{x}}_{\text{pr},ij}$, $\tilde{\boldsymbol{x}}_{\text{bl},ij}$ | |
| Total number of fields in population | $N$ | |
| Observed first stage sample vector | $\boldsymbol{s}_1$ | |
| Random first stage sample vector | $\boldsymbol{S}_1$ | |
| Subscript for sampled fields | $i = 1, \ldots, n$ | |
| Subscript for sampled points | $j = 1, \ldots, m_i$ | |
| Subscript for MC draw | $\ell = 1, \ldots, L$ | |

Table 8: Notation common to all sampling designs

| Quantity | Notation | Units |
|---|---|---|
| Predictive random variables | | |
| Total | $\tilde{\tau}$ | $tCO_2e$ |
| Areal mean | $\tilde{\mu}$ | $tCO_2e$/acre |
| Biogeochemical model parameters | $\boldsymbol{\theta}$ | |
| Draws from posterior predictive distribution | | |
| Total | $\tilde{\tau}_\ell$ | $tCO_2e$ |
| Areal mean | $\tilde{\mu}_\ell$ | $tCO_2e$/acre |
| Biogeochemical model parameters | $\boldsymbol{\theta}_\ell$ | |
| Estimated values | | |
| Total | $\hat{\tau}$ | $tCO_2e$ |
| Areal mean | $\hat{\mu}$ | $tCO_2e$/acre |
| True population values | | |
| Total | $\tau$ | $tCO_2e$ |
| Areal mean | $\mu$ | $tCO_2e$/acre |
| Subscript for pre-strata | $g = 1, \ldots, G$ | |
| Subscript for post-strata | $h = 1, \ldots, H$ | |

# C Technical background on Bayesian calibration of biogeochemical models

This appendix provides additional technical background to the overview in Section 6.1 and sets the stage for Appendices D and E. Following the notation of Kennedy and O'Hagan [31], let

$\boldsymbol{x}_i, z_i$, $i = 1, \ldots, n_{\text{cal}}$ be the covariates (e.g., initial %SOC, indicator for no-tillage) and measured outcomes (e.g., %SOC one year after making a practice change as measured by soil samples and dry combustion testing) respectively in the calibration data set. Let $\eta_i(\boldsymbol{\theta}) = \eta(\boldsymbol{x}_i, \boldsymbol{\theta})$ be output from the biogeochemical model for covariates $\boldsymbol{x}_i$ and parameters $\boldsymbol{\theta}$. Following Kennedy and O'Hagan [31] and Bayarri et al. [38], we assume that

- The calibration outcome data $z$ are obtained with measurement error, i.e.,

$$z_i = \zeta(\boldsymbol{x}_i) + \epsilon_i \tag{59}$$

  where $\zeta(\boldsymbol{x}_i)$ is the true value and $\epsilon_i \sim N(0, \sigma^2_\epsilon)$.

- The true value is equal to the model prediction plus a model inadequacy term

$$\zeta(\boldsymbol{x}_i) = \eta(\boldsymbol{x}_i, \boldsymbol{\theta}) + \delta(\boldsymbol{x}_i) \tag{60}$$

  where $\delta(\boldsymbol{x}_i) \sim N(B(\boldsymbol{x}_i), \sigma^2_\delta)$, $B(\boldsymbol{x}_i)$ is model bias, and $\delta(\boldsymbol{x}_i)$ is independent of the model output and measurement error $\epsilon_i$. (Kennedy and O'Hagan [31] also include a regression parameter $\rho$ that multiplies $\eta(\boldsymbol{x}_i, \boldsymbol{\theta})$ which we set to 1.)

Combining (59) and (60), it follows that $z_i = \eta(\boldsymbol{x}_i, \boldsymbol{\theta}) + \delta(\boldsymbol{x}_i) + \epsilon_i$. However, for biogeochemical models it is frequently helpful to log-transform the observations and model predictions [39] to obtain

$$\log(z_i) = \log\left(\eta(\boldsymbol{x}_i, \boldsymbol{\theta})\right) + \delta(\boldsymbol{x}_i) + \epsilon_i. \tag{61}$$

Let $\log(\boldsymbol{z}) = (\log(z_1), \ldots, \log(z_{n_{\text{cal}}}))$ and $\log(\boldsymbol{\eta}(\boldsymbol{\theta})) = (\log(\eta_1(\boldsymbol{\theta})), \ldots, \log(\eta_{n_{\text{cal}}}(\boldsymbol{\theta})))$. Following Gurung et al. [39], we assume the bias of the model inadequacy is zero, i.e., $B(\boldsymbol{x}) = 0$, and we log transform the observations and model predictions so that $\log(\boldsymbol{z}) \sim N(\log(\boldsymbol{\eta}(\boldsymbol{\theta})), \Sigma(\boldsymbol{\alpha}))$ where $\Sigma(\boldsymbol{\alpha})$ is the covariance matrix with parameters $\boldsymbol{\alpha}$. The covariance matrix $\Sigma(\boldsymbol{\alpha})$ includes variance due to model inadequacy $\sigma^2_\delta$ and measurement error $\sigma^2_\epsilon$.

Specifying priors $p(\boldsymbol{\theta}, \boldsymbol{\alpha})$ and the structure of the covariance matrix are an important part of the calibration process, though in this document we are exclusively focused on using the posterior $p(\boldsymbol{\theta}, \boldsymbol{\alpha}|\boldsymbol{z})$ to predict GHG emission reductions/removals at new fields and points outside of the calibration data set. Following conventions in Bayesian statistics, we use tildes to denote unobserved outcomes and observed covariates at points outside the calibration data set. Unobserved outcomes are said to follow a posterior predictive distribution. Please see Appendix B for additional details.

Let $(\boldsymbol{\theta}_\ell, \boldsymbol{\alpha}_\ell)$, $\ell = 1, \ldots, L$ be Monte Carlo (MC) draws from the posterior $p(\boldsymbol{\theta}, \boldsymbol{\alpha}|\boldsymbol{z})$, and let $\tilde{z}_{\text{pr},ij\ell}$ and $\tilde{z}_{\text{bl},ij\ell}$ be the corresponding draw from the posterior predictive distribution of GHG emissions under project and baseline scenarios[13] (recall Section 1.1) for field $i = 1, \ldots, n$ and point $j = 1, \ldots, m_i$:

- $p(\log(\tilde{z}_{\text{pr},ij})|\tilde{\boldsymbol{x}}_{\text{pr},ij}, \boldsymbol{z}) = \int p(\log(\tilde{z}_{ij})|\tilde{\boldsymbol{x}}_{\text{pr},ij}, \boldsymbol{\theta}, \boldsymbol{\alpha}) p(\boldsymbol{\theta}, \boldsymbol{\alpha}|\boldsymbol{z}) d\boldsymbol{\theta} d\boldsymbol{\alpha}$
- $p(\log(\tilde{z}_{\text{bl},ij})|\tilde{\boldsymbol{x}}_{\text{bl},ij}, \boldsymbol{z}) = \int p(\log(\tilde{z}_{ij})|\tilde{\boldsymbol{x}}_{\text{bl},ij}, \boldsymbol{\theta}, \boldsymbol{\alpha}) p(\boldsymbol{\theta}, \boldsymbol{\alpha}|\boldsymbol{z}) d\boldsymbol{\theta} d\boldsymbol{\alpha}$.

We are interested in the GHG emission reductions and removals at field $i$ point $j$, denoted by $\tilde{y}_{ij} = \tilde{z}_{\text{pr},ij} - \tilde{z}_{\text{bl},ij}$. The $\ell^{th}$ MC draw from the posterior predictive distribution $p(\tilde{y}_{ij}|\boldsymbol{z})$ is $\tilde{y}_{ij\ell} = \tilde{z}_{\text{pr},ij\ell} - \tilde{z}_{\text{bl},ij\ell}$.

[13] Per the SEP [1], the baseline is an average of several different crop patterns. The posterior draw $\tilde{z}_{\text{bl},ij\ell}$ represents an average of baseline scenarios.

# D Derivation of estimators based on Bayesian-calibrated model predictions

Per Table 8, let $\tau$ be the true population total of GHG emission reductions/removals in the form of $tCO_2e$ of $CO_2$, $N_2O$, and $CH_4$ molecules that are not in the atmosphere due to the land management practice changes of growers participating in the SEP project. In this appendix, we derive estimators of the population total $\tau$ and associated variance. These results extend the work of Del Grosso et al. [40] to different sampling designs common in SEP projects, and they propose a bootstrap approach for situations in which the approach of Del Grosso et al. [40] cannot be used.

The examples in the main text are applications of the derivations in this appendix. The derivations in Appendices D.1 and D.2 give the variance estimator of Del Grosso et al. [40] and are very similar with slight differences to account for the single-stage versus two-stage designs. The variance estimators in Appendices D.1 and D.2 can be used in unadjusted analyses and in post-stratified analyses where primary sampling units are not split across post-strata. The approach in Appendix D.3 is an alternative to those in Appendices D.1 and D.2, and can also be used in situations where primary sampling units are split across post-strata. However, the approach in Appendix D.3 is also more computationally intensive.

Please see Appendix B for notation.

## D.1 SRSWR Design

This appendix uses notation in Tables 6 and 8. In a simple random sample with replacement (SRSWR) design, the population of primary units (points) $j = 1, \ldots, N$ are known at the time of sampling and we treat the covariates as fixed. While the biogeochemical model predictions at a given point are random variables, the sampling of fields and points is conducted under a design-based framework.

We can construct a posterior predictive random variable of the total as

$$\tilde{\tau} = \frac{A}{n} \sum_{j=1}^{n} \tilde{Y}_j \tag{62}$$

Let $\boldsymbol{z} = (z_1, \ldots, z_{n_{\text{cal}}})$ be the calibration data, and let $\boldsymbol{s} = (s_1, \ldots, s_N)$ be a sample vector where $s_i$ is the number of times unit $i$ was selected. The posterior expectation of (62) conditional on the calibration data $\boldsymbol{z}$ and the sample $\boldsymbol{s}$ is[14]

$$\mathbb{E}[\tilde{\tau}|\boldsymbol{s}, \boldsymbol{z}] = \frac{A}{n} \sum_{j=1}^{n} \mathbb{E}\left[\tilde{Y}_j|\boldsymbol{z}\right]. \tag{63}$$

An unbiased estimator of (63) is

$$\hat{\tau} = \widehat{\mathbb{E}}[\tilde{\tau}|\boldsymbol{s}, \boldsymbol{z}] = \frac{A}{n} \sum_{j=1}^{n} \bar{\tilde{y}}_j \tag{64}$$

where $\bar{\tilde{y}}_j = L^{-1} \sum_{\ell=1}^{L} \tilde{y}_{j\ell}$.

[14] We explicitly condition on the sample $\boldsymbol{s}$ in this appendix to clarify derivations. In the examples of section 6.2 we do not explicitly condition on $\boldsymbol{s}$ to avoid confusion, though the conditioning is implicit.

If the posterior model predictions are unbiased for the true GHG emission reductions/removals and there is no bias in MC estimation, then by the properties of simple random sampling with replacement [e.g., see 25], $\hat{\tau}$ is design-unbiased for the true population total $\tau$ (please see note 4). In other words,

$$\mathbb{E}_{\boldsymbol{S}}[\hat{\tau}|\boldsymbol{z}] = \mathbb{E}_{\boldsymbol{S}}\left[\hat{\mathbb{E}}\left[\tilde{\tau}|\boldsymbol{s},\boldsymbol{z}\right]|\boldsymbol{z}\right] = \frac{A}{N}\sum_{i=1}^{N}\mathbb{E}[\tilde{Y}_i|\boldsymbol{z}] \qquad \text{(If there is no MC bias)} \tag{65a}$$

$$= \tau \qquad \text{(If } \mathbb{E}[\tilde{Y}_i|\boldsymbol{z}] \text{ is unbiased for truth)} \tag{65b}$$

where the $\boldsymbol{S}$ subscript on $\mathbb{E}_{\boldsymbol{S}}$ indicates that the expectation is taken with respect to all possible SRSWR samples of $n$ points. All expectations and variances in this section without a subscript are with respect to the posterior predictive distribution of $\tilde{\tau}$.

**Note 4: Design-unbiasedness of posterior model predictions**

As we note above, $\hat{\tau}$ is design-unbiased for the true population total if the posterior predictions are unbiased and there is no bias in MC estimation. To clarify this point, we note that $\mathbb{E}[\tilde{Y}_i|\boldsymbol{s},\boldsymbol{z}]$ is a function of the sample $\boldsymbol{s}$ and calibration data $\boldsymbol{z}$. While $\tilde{Y}_i$ is random, $\mathbb{E}[\tilde{Y}_i|\boldsymbol{s},\boldsymbol{z}]$ is fixed for fixed $\boldsymbol{s}$ and $\boldsymbol{z}$. Said another way, for fixed calibration data $\boldsymbol{z}$, $\mathbb{E}[\tilde{Y}_i|\boldsymbol{s},\boldsymbol{z}]$ can be analyzed as fixed measurements in a design-based framework in which the only random quantity is the sample $\boldsymbol{s}$.

Because the sample mean is design-unbiased under SRSWR, $\hat{\tau}$ is also design-unbiased for the population total of expected posterior predictions $(A/N)\sum_{i=1}^{N}\mathbb{E}[\tilde{Y}_i|\boldsymbol{z}]$ if there is no bias in the MC estimation. If in addition the posterior expectations $\mathbb{E}[\tilde{Y}_i|\boldsymbol{z}]$ are unbiased for the true GHG emissions reductions/removals, then $\hat{\tau}$ is also design-unbiased for the true population total $\tau$. This highlights the importance of calibrating and validating biogeochemical models to ensure minimal to no bias in posterior predictions.

While design-based principles can be applied to estimators of the total (and mean) to demonstrate unbiasedness under certain conditions, in practice the unbiasedness of $\mathbb{E}[\tilde{Y}|\boldsymbol{z}]$ may not hold exactly and design-based estimators of variance do not capture this source of uncertainty. As a consequence, a design-based estimator of variance tends to underestimate variability in this setting, leading to confidence intervals with below nominal coverage probability for the true GHG emission reductions/removals $\tau$. The variance estimator described below corrects this problem.

Let

$$\tilde{\tau}_\ell = \frac{A}{n}\sum_{j=1}^{n}\tilde{y}_{j\ell} \tag{66}$$

be the $\ell^{th}$ draw from the posterior predictive distribution of $p(\tilde{\tau}|\boldsymbol{s},\boldsymbol{z})$. An unbiased estimate of the posterior variance of $\tilde{\tau}$ conditional on the sample $\boldsymbol{s}$ is

$$\widehat{\mathrm{Var}}\left(\tilde{\tau}|\boldsymbol{s},\boldsymbol{z}\right) = \frac{1}{L-1}\sum_{\ell=1}^{L}(\tilde{\tau}_\ell - \hat{\tau})^2 \tag{67}$$

where $\hat{\tau} = L^{-1}\sum_{\ell=1}^{L}\tilde{\tau}_\ell = (A/n)\sum_{j=1}^{n}\bar{\tilde{y}}_j$.

However, to make inference about the population total we must also account for sampling variability. In other words, we want to estimate $\text{Var}(\tilde{\tau}|\boldsymbol{z})$ which can be decomposed as

$$\text{Var}\left(\tilde{\tau}|\boldsymbol{z}\right) = \text{Var}_{\boldsymbol{S}}\left(\mathbb{E}\left[\tilde{\tau}|\boldsymbol{s},\boldsymbol{z}\right]|\boldsymbol{z}\right) + \mathbb{E}_{\boldsymbol{S}}\left[\text{Var}\left(\tilde{\tau}|\boldsymbol{s},\boldsymbol{z}\right)|\boldsymbol{z}\right]. \tag{68}$$

An unbiased estimator of the expectation inside the first term of (68) is given by (64), and an unbiased estimator of the sampling variance of (64) is

$$\widehat{\text{Var}}_{\boldsymbol{S}}\left(\frac{A}{n}\sum_{j=1}^{n}\bar{\tilde{y}}_j\right) = \frac{A^2}{n(n-1)}\sum_{j=1}^{n}(\bar{\tilde{y}}_j - \hat{\mu})^2 \tag{69}$$

where $\hat{\mu} = \hat{\tau}/A = n^{-1}\sum_{j=1}^{n}\bar{\tilde{y}}_i$.

An unbiased estimator of the posterior variance inside the second term of (68) is given by (67). Before taking the expectation of (67), we note that for fixed calibration data $\boldsymbol{z}$ the posterior variance estimator is solely a function of the sample $\boldsymbol{s}$, and we let $\hat{v}(\boldsymbol{s}) = \widehat{\text{Var}}\left(\tilde{\tau}|\boldsymbol{s},\boldsymbol{z}\right)$. Then

$$\mathbb{E}_{\boldsymbol{S}}[\hat{v}(\boldsymbol{s})] = \sum_{\boldsymbol{s}}\hat{v}(\boldsymbol{s})p(\boldsymbol{s}) \tag{70}$$

where the sum is over all SRSWR samples of $n$ points, and $p(\boldsymbol{s})$ is the probability of drawing sample $\boldsymbol{s}$. To the best of our knowledge, there is not an analytic solution to (70) for this scenario in which $\hat{v}(\boldsymbol{s})$ is the variance of a posterior predictive distribution. However, following Del Grosso et al. [40], we estimate (70) using our single observed sample as

$$\widehat{\mathbb{E}}_{\boldsymbol{S}}\left[\widehat{\text{Var}}\left(\tilde{\tau}|\boldsymbol{s},\boldsymbol{z}\right)|\boldsymbol{z}\right] = \widehat{\text{Var}}\left(\tilde{\tau}|\boldsymbol{s},\boldsymbol{z}\right). \tag{71}$$

To support the use of (71), Appendix E derives probabilistic bounds on the magnitude of error and shows that the error goes to zero in probability as the sample size $n$ gets large. In particular, $|\widehat{\text{Var}}(\tilde{\tau}|\boldsymbol{s},\boldsymbol{z}) - \mathbb{E}_{\boldsymbol{S}}[\widehat{\text{Var}}(\tilde{\tau}|\boldsymbol{s},\boldsymbol{z})|\boldsymbol{z}]| = o_p(1)$.

Putting the results in (69) and (71) together,

$$\begin{aligned}\widehat{\text{Var}}(\tilde{\tau}|\boldsymbol{z}) &= \widehat{\text{Var}}_{\boldsymbol{S}}\left(\widehat{\mathbb{E}}\left[\tilde{\tau}|\boldsymbol{s},\boldsymbol{z}\right]|\boldsymbol{z}\right) + \widehat{\mathbb{E}}_{\boldsymbol{S}}\left[\widehat{\text{Var}}\left(\tilde{\tau}|\boldsymbol{s},\boldsymbol{z}\right)|\boldsymbol{z}\right] \\ &= \frac{A^2}{n(n-1)}\sum_{j=1}^{n}(\bar{\tilde{y}}_j - \hat{\mu})^2 + \frac{1}{L-1}\sum_{\ell=1}^{L}(\tilde{\tau}_\ell - \hat{\tau})^2. \end{aligned} \tag{72}$$

The first term of (72) is equivalent to [40, Equation 5] but for sampling with replacement, and the second term of (72) is equivalent to [40, Equation 3].

To get the variance of the areal mean $\hat{\mu}$, both terms of (72) are divided by $A^2$.

## D.2 PPSWR-SRSWR Design

This appendix uses notation in Tables 7 and 8. Similar to the one-stage SRSWR design, in a probability proportional to size with replacement followed by simple random sample with replacement (PPSWR-SRSWR) design, the population of primary units (fields) and their areas $A_i$, $i = 1, \ldots, N$ are known at the time of sampling and we treat the covariates as fixed. While the biogeochemical model predictions at a given point are random variables, the sampling of fields and points is conducted under a design-based framework.

The probability of selecting field $i$ in each draw is $p_i = A_i/A$ where $A = \sum_{i=1}^{n} A_i$, and we can construct a Hansen-Hurwitz type posterior predictive random variable of the total as

$$\tilde{\tau} = \frac{1}{n}\sum_{i=1}^{n}\frac{A_i}{p_i}\tilde{Y}_i = \frac{A}{n}\sum_{i=1}^{n}\tilde{Y}_i \tag{73}$$

where $\tilde{Y}_i = m_i^{-1}\sum_{j=1}^{m_i}\tilde{Y}_{ij}$.

Let $\boldsymbol{z} = (z_1, \ldots, z_{n_{\text{cal}}})$ be the calibration data, and let $\boldsymbol{s}_1 = (s_{11}, \ldots, s_{1N})$ where $s_{1i}$ is the number of times field $i$ is selected in the first stage of sampling. The posterior expectation of (73) conditional on the first stage sample $\boldsymbol{s}_1$ is[15]

$$\mathbb{E}[\tilde{\tau}|\boldsymbol{s}_1, \boldsymbol{z}] = \frac{A}{n}\sum_{i=1}^{n}\mathbb{E}\left[\tilde{Y}_i|\boldsymbol{z}\right]. \tag{74}$$

An unbiased estimator of (74) is

$$\hat{\tau} = \widehat{\mathbb{E}}[\tilde{\tau}|\boldsymbol{s}_1, \boldsymbol{z}] = \frac{A}{n}\sum_{i=1}^{n}\bar{\tilde{y}}_i \tag{75}$$

where $\bar{\tilde{y}}_i = m_i^{-1}\sum_{j=1}^{m_i}\bar{\tilde{y}}_{ij}$, and $\bar{\tilde{y}}_{ij} = L^{-1}\sum_{\ell=1}^{L}\tilde{y}_{ij\ell}$.

Similar to Appendix D.1, if the posterior predictions are unbiased for the true GHG emission reductions/removals and there is no MC bias, then by the properties of the Hansen-Hurwitz estimator in two-stage designs [e.g., see 25, Ch. 13.5], $\hat{\tau}$ is design-unbiased for the true population total $\tau$ (see note 4). In other words,

$$\mathbb{E}_{\boldsymbol{S}_1}[\hat{\tau}|\boldsymbol{z}] = \mathbb{E}_{\boldsymbol{S}_1}\left[\hat{\mathbb{E}}\left[\tilde{\tau}|\boldsymbol{s}_1, \boldsymbol{z}\right]|\boldsymbol{z}\right] = \sum_{i=1}^{N} A_i\mathbb{E}[\tilde{Y}_i|\boldsymbol{z}] \qquad \text{(If there is no MC bias)} \tag{76a}$$

$$= \tau \qquad \text{(If } \mathbb{E}[\tilde{Y}_i|\boldsymbol{z}] \text{ is unbiased for truth)} \tag{76b}$$

where the $\boldsymbol{s}_1$ subscript on $\mathbb{E}_{\boldsymbol{S}_1}$ indicates that the expectation is taken with respect to all possible first stage samples of $n$ primary units. As in Appendix D.1, all expectations and variances in this section without a subscript are with respect to the posterior predictive distribution of $\tilde{\tau}$.

Let

$$\tilde{\tau}_\ell = \frac{A}{n}\sum_{i=1}^{n}\left(\frac{1}{m_i}\sum_{j=1}^{m_i}\tilde{y}_{ij\ell}\right) \tag{77}$$

be the $\ell^{th}$ draw from the posterior predictive distribution of $p(\tilde{\tau}|\boldsymbol{s}_1, \boldsymbol{z})$. An unbiased estimate of the posterior variance of $\tilde{\tau}$ conditional on the first stage sample $\boldsymbol{s}_1$ is

$$\widehat{\text{Var}}\left(\tilde{\tau}|\boldsymbol{s}_1, \boldsymbol{z}\right) = \frac{1}{L-1}\sum_{\ell=1}^{L}(\tilde{\tau}_\ell - \hat{\tau})^2 \tag{78}$$

[15] We explicitly condition on the first stage sample $\boldsymbol{s}_1$ in this appendix to clarify derivations. In the examples of section 6.2 we do not explicitly condition on $\boldsymbol{s}_1$ to avoid confusion, though the conditioning is implicit.

where $\hat{\tau} = L^{-1}\sum_{\ell=1}^{L}\tilde{\tau}_\ell = (A/n)\sum_{i=1}^{n}\bar{\tilde{y}}_i$.

However, to make inference about the population total we must also account for sampling variability. In other words, we want to estimate $\text{Var}(\tilde{\tau}|\boldsymbol{z})$ which can be decomposed as

$$\text{Var}\left(\tilde{\tau}|\boldsymbol{z}\right) = \text{Var}_{\boldsymbol{S}_1}\left(\mathbb{E}\left[\tilde{\tau}|\boldsymbol{s}_1,\boldsymbol{z}\right]|\boldsymbol{z}\right) + \mathbb{E}_{\boldsymbol{S}_1}\left[\text{Var}\left(\tilde{\tau}|\boldsymbol{s}_1,\boldsymbol{z}\right)|\boldsymbol{z}\right]. \tag{79}$$

An unbiased estimator of the expectation inside the first term of (79) is given by (75), and an unbiased estimator of the sampling variance of (75) is

$$\widehat{\text{Var}}_{\boldsymbol{S}_1}\left(\frac{A}{n}\sum_{i=1}^{n}\bar{\tilde{y}}_i\right) = \frac{A^2}{n(n-1)}\sum_{i=1}^{n}(\bar{\tilde{y}}_i - \hat{\mu})^2 \tag{80}$$

where $\hat{\mu} = \hat{\tau}/A = n^{-1}\sum_{i=1}^{n}\bar{\tilde{y}}_i$.

An unbiased estimator of the posterior variance inside the second term of (79) is given by (78). By the same argument as in Appendix D.1 we estimate the expected posterior variance as

$$\widehat{\mathbb{E}}_{\boldsymbol{S}_1}\left[\widehat{\text{Var}}\left(\tilde{\tau}|\boldsymbol{s}_1,\boldsymbol{z}\right)|\boldsymbol{z}\right] = \widehat{\text{Var}}\left(\tilde{\tau}|\boldsymbol{s}_1,\boldsymbol{z}\right). \tag{81}$$

Putting the results in (80) and (81) together,

$$\begin{aligned}\widehat{\text{Var}}(\tilde{\tau}|\boldsymbol{z}) &= \widehat{\text{Var}}_{\boldsymbol{S}_1}\left(\widehat{\mathbb{E}}\left[\tilde{\tau}|\boldsymbol{s}_1,\boldsymbol{z}\right]|\boldsymbol{z}\right) + \widehat{\mathbb{E}}_{\boldsymbol{S}_1}\left[\widehat{\text{Var}}\left(\tilde{\tau}|\boldsymbol{s}_1,\boldsymbol{z}\right)|\boldsymbol{z}\right] \\ &= \frac{A^2}{n(n-1)}\sum_{i=1}^{n}(\bar{\tilde{y}}_i - \hat{\mu})^2 + \frac{1}{L-1}\sum_{\ell=1}^{L}(\tilde{\tau}_\ell - \hat{\tau})^2. \end{aligned} \tag{82}$$

The first term of (82) is equivalent to [40, Equation 5] but for sampling primary units with replacement, and the second term of (82) is equivalent to [40, Equation 3].

To get the variance of the areal mean $\hat{\mu}$, both terms of (82) are divided by $A^2$.

### D.3 Novel approach for any design and post-stratification scheme

The approaches in Appendices D.1 and D.2 depend on being able to decompose the variance into sampling and modeling variability as in Equations (68) and (79). However, this is not possible in post-stratified analyses where primary sampling units are split across post-strata, such as in Examples 4 and 10 which post-stratify at the sub-field level (secondary sampling units) in PPSWR-SRSWR designs. The resampling approach described in this section is an alternative to the approaches in Appendices D.1 and D.2, and it can be applied in more general settings including the situation described above.

#### D.3.1 Description of the problem when post-stratifying at the sub-field level

Consider the setting of Example 10. As in Appendix D.1, we can construct a posterior predictive random variable of the population total in post-stratum $h$ as

$$\tilde{\tau}_h = \frac{A_h}{n_h}\sum_{j=1}^{n_h}\tilde{Y}_{hj} \tag{83}$$

and for the entire population as

$$\tilde{\tau}_{\text{ps}} = \sum_{h} \tilde{\tau}_h. \tag{84}$$

Let $\boldsymbol{z} = (z_1, \ldots, z_{n_{\text{cal}}})$ be the calibration data. The posterior expectation of (83) conditional on the calibration data $\boldsymbol{z}$ and vector of first stage sampling units $\boldsymbol{s}_1$ is

$$\mathbb{E}[\tilde{\tau}_h | \boldsymbol{s}_1, \boldsymbol{z}] = \frac{A}{n} \sum_{j=1}^{n} \mathbb{E}\left[\tilde{Y}_{hj} | \boldsymbol{z}\right] \tag{85}$$

and an unbiased estimator of (85) is

$$\hat{\tau}_h = \widehat{\mathbb{E}}[\tilde{\tau} | \boldsymbol{s}_1, \boldsymbol{z}] = \frac{A_h}{n_h} \sum_{j=1}^{n_h} \bar{\tilde{y}}_{hj} \tag{86}$$

where $\bar{\tilde{y}}_{hj} = L^{-1} \sum_{\ell=1}^{L} \tilde{y}_{hj\ell}$. An unbiased estimator of the population total is

$$\hat{\tau}_{\text{ps}} = \sum_{h=1}^{H} \hat{\tau}_h. \tag{87}$$

Let

$$\tilde{\tau}_{h\ell} = \frac{A_h}{n_h} \sum_{j=1}^{n_h} y_{hj\ell} \tag{88}$$

be the $\ell^{th}$ draw from the posterior predictive distribution of $p(\tilde{\tau} | \boldsymbol{s}_1, \boldsymbol{z})$ and let

$$\tilde{\tau}_{\text{ps},\ell} = \sum_{h=1}^{H} \tilde{\tau}_{h\ell}. \tag{89}$$

An unbiased estimate of the posterior variance of $\tilde{\tau}_{\text{ps}}$ conditional on the first-stage sample $\boldsymbol{s}_1$ is

$$\widehat{\text{Var}}\left(\tilde{\tau}_{\text{ps}} | \boldsymbol{s}_1, \boldsymbol{z}\right) = \frac{1}{L-1} \sum_{\ell=1}^{L} (\tilde{\tau}_{\text{ps},\ell} - \hat{\tau}_{\text{ps}})^2. \tag{90}$$

The calculation of variance (90) cannot be split across strata because the same MC draws can be used across strata, which can induce dependence.

At this point, the approaches in Appendices D.1 and D.2 would decompose the variance as

$$\begin{aligned}
\text{Var}\left(\tilde{\tau}_{\text{ps}} | \boldsymbol{z}\right) &= \text{Var}_{\boldsymbol{S}_1}\left(\mathbb{E}\left[\tilde{\tau}_{\text{ps}} | \boldsymbol{s}_1, \boldsymbol{z}\right] | \boldsymbol{z}\right) + \mathbb{E}_{\boldsymbol{S}_1}\left[\text{Var}\left(\tilde{\tau}_{\text{ps}} | \boldsymbol{s}_1, \boldsymbol{z}\right) | \boldsymbol{z}\right] && \text{(91a)} \\
&= \text{Var}_{\boldsymbol{S}_1}\left(\sum_{h=1}^{H} \mathbb{E}\left[\tilde{\tau}_h | \boldsymbol{s}_1, \boldsymbol{z}\right] | \boldsymbol{z}\right) + \mathbb{E}_{\boldsymbol{S}_1}\left[\text{Var}\left(\tilde{\tau}_{\text{ps}} | \boldsymbol{s}_1, \boldsymbol{z}\right) | \boldsymbol{z}\right] && \text{(91b)} \\
&= \sum_{h=1}^{H} \left\{\text{Var}_{\boldsymbol{S}_1}\left(\mathbb{E}\left[\tilde{\tau}_h | \boldsymbol{s}_1, \boldsymbol{z}\right] | \boldsymbol{z}\right)\right\} + \mathbb{E}_{\boldsymbol{S}_1}\left[\text{Var}\left(\tilde{\tau}_{\text{ps}} | \boldsymbol{s}_1, \boldsymbol{z}\right) | \boldsymbol{z}\right] && \text{(91c)}
\end{aligned}$$

where line (91c) follows because the conditional expectations are independent due to the sampling design.

However, while we can estimate the posterior expectation within each stratum conditional on $\boldsymbol{s}_1$ using Equation (86), we cannot estimate the outer variance in the first term of (91c). This is because the outer variance is an integral over the set of all samples of primary units, but the primary units are split across post-strata (in contrast, the inner expectation and variance are with respect to the posterior distribution, conditional on the observed sample), and there may be correlations between post-strata. More concretely, the estimator of the first term in (91c) would need to be of the same form as (80), which calculates the sampling variance between primary units. However, when the primary units are split across post-strata, it is not possible to use a calculation similar to (80) within post-strata, in part because correlations between post-strata must also be taken into account.

### D.3.2 Description of solution with bootstrap estimator

As in Example 4, we emulate the sampling mechanism to estimate variance, which can be done with bootstrap resampling. For each of $b = 1, \ldots, B$ resamples, a sample of $n$ of the observed primary units are resampled with replacement to form the $b^{\text{th}}$ bootstrap resample. For each bootstrap resample, let

$$\tilde{\tau}_{\text{ps},\ell}^{(b)} = \sum_{h=1}^{H} \frac{A_h}{n_h^{(b)}} \sum_{j=1}^{n_h^{(b)}} \tilde{y}_{hj\ell}^{(b)} \tag{92}$$

be the $\ell^{th}$ draw from the posterior predictive distribution of the population total. The $b^{\text{th}}$ bootstrap post-stratified estimate of the total is

$$\hat{\tau}_{\text{ps}}^{(b)} = \frac{1}{L} \sum_{\ell=1}^{L} \tilde{\tau}_{\text{ps},\ell}^{(b)} \tag{93a}$$

$$= \sum_{h=1}^{H} \frac{A_h}{n_h^{(b)}} \sum_{j=1}^{n_h^{(b)}} \bar{\tilde{y}}_{hj}^{(b)} \tag{93b}$$

where $\bar{\tilde{y}}_{hj}^{(b)} = L^{-1} \sum_{\ell=1}^{L} \tilde{y}_{hj\ell}^{(b)}$.

The variance estimator for the $b^{th}$ bootstrap resample is

$$v^{(b)} = \frac{1}{L-1} \sum_{\ell=1}^{L} \left( \tilde{\tau}_{\text{ps},\ell}^{(b)} - \hat{\tau}_{\text{ps}}^{(b)} \right)^2. \tag{94}$$

Let

$$\hat{\tau}_{\text{ps,boot}} = \frac{1}{B} \sum_{b=1}^{B} \hat{\tau}_{\text{ps}}^{(b)} \tag{95}$$

be the mean of the bootstrap totals.

The variance of $\hat{\tau}_{\text{ps,boot}}$ can be decomposed into between- and within-bootstrap variance components (i.e., sampling and model variance, respectively) as

$$\text{Var}\left(\hat{\tau}_{\text{ps,boot}} | \boldsymbol{z}\right) = \text{Var}\left(\mathbb{E}\left[\hat{\tau}_{\text{ps,boot}} | b, \boldsymbol{z}\right] | \boldsymbol{z}\right) + \mathbb{E}\left[\text{Var}\left(\hat{\tau}_{\text{ps,boot}} | b, \boldsymbol{z}\right) | \boldsymbol{z}\right] \tag{96a}$$

which can be estimated as

$$\widehat{\text{Var}}(\hat{\tau}_{\text{ps,boot}}|\boldsymbol{z}) = v_{\text{sample}} + v_{\text{model}} \tag{97}$$

where

$$v_{\text{sample}} = \frac{1}{B-1}\sum_{b=1}^{B}(\hat{\tau}_{\text{ps}}^{(b)} - \hat{\tau}_{\text{ps,boot}})^2 \tag{98a}$$

$$v_{\text{model}} = \frac{1}{B}\sum_{b=1}^{B} v^{(b)}. \tag{98b}$$

Note that (97) accounts for all correlations between post-strata, because (98a) and (98b) are computed with the post-stratified totals.

While we derive this bootstrap estimator in the context of a post-stratification scheme that splits primary sampling units across post-strata, it can also be adapted to simpler settings as an alternative to the calculations in Appendices D.1 and D.2.

# E Error bound on the estimate of model variance in Equation (71)

In this section, we derive probabilistic bounds for $|\hat{v}(\boldsymbol{s}) - \mathbb{E}_{\boldsymbol{S}}[\hat{v}(\boldsymbol{s})]|$ where $\hat{v}(\boldsymbol{s}) = \widehat{\text{Var}}(\tilde{\tau}|\boldsymbol{s}, \boldsymbol{z})$ to support the use of $\hat{v}(\boldsymbol{s})$ as an estimate of $\mathbb{E}_{\boldsymbol{S}}[\hat{v}(\boldsymbol{s})]$ in (71) and (81) as is done by Del Grosso et al. [40]. The particular bound in Lemma E.1 is specific to (71), but adaptations to (81) are straightforward and the rate of convergence to zero is the same. This section uses notation in Tables 6 and 8 in Appendix B.

To the best of our knowledge, existing results on the convergence of the sample variance focus on the case where the sample is composed of identically distributed observations and the sample size becomes large (e.g., see [41, Theorem 5.3.1]. However, in our setting the number of observations $L$ is typically fixed and the posterior predictive quantities $\tilde{\tau}_\ell$, $\ell = 1, \ldots, L$ used to compute the sample variance are themselves sample totals. As the sample size $n$ becomes large, the $\tilde{\tau}_\ell$ converge in probability to their population quantities, and it follows that the sample variance of the $\tilde{\tau}_\ell$ converges in probability to the population variance. Lemma E.1 makes this precise and provides a probabilistic bound on the error. To the best of our knowledge, these bounds have not been derived elsewhere.

In Lemma E.1 and its proof, we denote errors as $\delta$ and $d$ and bounds as $\epsilon$. The use of these symbols in this section is distinct from their use in other parts of this document. In particular, there is no connection between the errors $\delta$ and $d$ in this section and the model inadequacy term in Appendix C or standardized difference in Section 3.2, or between the bound $\epsilon$ in this section and the measurement error in Appendix C.

Let $\tilde{\tau}_\ell^{\text{pop}} = \mathbb{E}_{\boldsymbol{S}}[\tilde{\tau}_\ell|\boldsymbol{z}]$ and $\hat{\tau}^{\text{pop}} = L^{-1}\sum_{\ell=1}^{L}\tilde{\tau}_\ell^{\text{pop}}$ be the true posterior population totals. Also, let $z_\alpha = \Phi^{-1}(\alpha)$ where $\Phi$ is the standard normal distribution. For example, for $\alpha = 0.05$, $z_{1-\alpha/2} = 1.96$. Also, let $\sigma_\ell^2 = \text{Var}_{\boldsymbol{S}}(\tilde{y}_{1\ell})$, which can be estimated as $\hat{\sigma}_\ell^2 = (n-1)^{-1}\sum_{j=1}^{n}(\tilde{y}_{j\ell} - \bar{\tilde{y}}_{\cdot\ell})^2$ where $\bar{\tilde{y}}_{\cdot\ell} = n^{-1}\sum_{j=1}^{n} y_{j\ell}$. As in Appendix D.1, let $\hat{v}(\boldsymbol{s}) = \widehat{\text{Var}}(\tilde{\tau}|\boldsymbol{s}, \boldsymbol{z})$ be the estimated posterior variance of the population total $\tilde{\tau}$. We note that a two-sided $1-\alpha/2$ confidence interval for $\tilde{\tau}_\ell$ is $\tilde{\tau}_\ell \pm z_{1-\alpha/2}A\sigma_\ell/\sqrt{n}$, which we use in Lemma E.1 to bound errors in probability. We also note that quantiles from a

$t$-distribution could be used instead of quantiles from a normal distribution, particularly for small sample sizes (e.g., $n < 50$).

**Lemma E.1** (Probabilistic bound on the error of (71) under SRSWR)**.** *Conditional on posterior predictive MC draws $\tilde{y}_{j\ell}$, $j = 1, \ldots, n$, $\ell = 1, \ldots, L$, with probability $1 - \alpha/2$*

$$|\hat{v}(\boldsymbol{s}) - \mathbb{E}_{\boldsymbol{S}}[\hat{v}(\boldsymbol{s})]| \leq \frac{1}{L-1} \sum_{\ell=1}^{L} \left\{ 2|\tilde{\tau}_\ell^{pop} - \hat{\tau}^{pop}|\epsilon_\ell + \epsilon_\ell^2 \right\}$$

*where*

$$\epsilon_\ell = A \frac{(L-1) z_{1-\alpha/2} \sigma_\ell + \sum_{k \neq \ell} z_{1-\alpha/2} \sigma_k}{L\sqrt{n}}.$$

*Proof.* Let

$$\hat{v}(\boldsymbol{s}) = \frac{1}{L-1} \sum_{\ell=1}^{L} (\tilde{\tau}_\ell - \hat{\tau})^2 \tag{99a}$$

$$= \frac{1}{L-1} \sum_{\ell=1}^{L} U_\ell^2 \tag{99b}$$

where

$$U_\ell = \tilde{\tau}_\ell - \hat{\tau} \tag{100a}$$

$$= \tilde{\tau}_\ell^{\text{pop}} - \hat{\tau}^{\text{pop}} + (\tilde{\tau}_\ell - \tilde{\tau}_\ell^{\text{pop}}) + (\hat{\tau}^{\text{pop}} - \hat{\tau}). \tag{100b}$$

Also let

$$\delta_\ell = \tilde{\tau}_\ell - \tilde{\tau}_\ell^{\text{pop}}. \tag{101}$$

Then for $n$ sufficiently large that a normal approximation holds, with probability $1 - \alpha$

$$|\delta_\ell| < z_{1-\alpha/2} A \sigma_\ell / \sqrt{n}. \tag{102}$$

In addition,

$$\hat{\tau} = \frac{1}{L} \sum_{\ell=1}^{L} \tilde{\tau}_\ell \tag{103a}$$

$$= \frac{1}{L} \sum_{\ell=1}^{L} (\tilde{\tau}_\ell^{\text{pop}} + \tilde{\tau}_\ell - \tilde{\tau}_\ell^{\text{pop}}) \tag{103b}$$

$$= \hat{\tau}^{\text{pop}} + \frac{1}{L} \sum_{\ell=1}^{L} \delta_\ell. \tag{103c}$$

Therefore,

$$U_\ell = \tilde{\tau}_\ell^{\text{pop}} - \hat{\tau}^{\text{pop}} + \delta_\ell - \frac{1}{L} \sum_{k=1}^{L} \delta_k \tag{104a}$$

$$= \tilde{\tau}_\ell^{\text{pop}} - \hat{\tau}^{\text{pop}} + \frac{L-1}{L} \delta_\ell - \frac{1}{L} \sum_{k \neq \ell} \delta_k. \tag{104b}$$

Let

$$d_\ell = \frac{L-1}{L}\delta_\ell - \frac{1}{L}\sum_{k\neq\ell}\delta_k \tag{105}$$

and note that with probability $1-\alpha/2$,

$$|d_\ell| \leq \frac{L-1}{L} z_{1-\alpha/2}\frac{A\sigma_\ell}{\sqrt{n}} + \frac{1}{L}\sum_{k\neq\ell} z_{1-\alpha/2}\frac{A\sigma_k}{\sqrt{n}} \tag{106a}$$

$$= A\frac{(L-1)z_{1-\alpha/2}\sigma_\ell + \sum_{k\neq\ell} z_{1-\alpha/2}\sigma_k}{L\sqrt{n}} \tag{106b}$$

$$\equiv \epsilon_\ell. \tag{106c}$$

Now,

$$\hat{v}(\boldsymbol{s}) = \frac{1}{L-1}\sum_{\ell=1}^{L} U_\ell^2 \tag{107a}$$

$$= \frac{1}{L-1}\sum_{\ell=1}^{L}\left\{(\tilde{\tau}_\ell^{\text{pop}} - \hat{\tau}^{\text{pop}})^2 + 2(\tilde{\tau}_\ell^{\text{pop}} - \hat{\tau}^{\text{pop}})d_\ell + d_\ell^2\right\} \tag{107b}$$

$$= \mathbb{E}_{\boldsymbol{S}}[\hat{v}(\boldsymbol{s})] + \frac{1}{L-1}\sum_{\ell=1}^{L}\left\{2(\tilde{\tau}_\ell^{\text{pop}} - \hat{\tau}^{\text{pop}})d_\ell + d_\ell^2\right\} \tag{107c}$$

so

$$\hat{v}(\boldsymbol{s}) - \mathbb{E}_{\boldsymbol{S}}[\hat{v}(\boldsymbol{s})] = \frac{1}{L-1}\sum_{\ell=1}^{L}\left\{2(\tilde{\tau}_\ell^{\text{pop}} - \hat{\tau}^{\text{pop}})d_\ell + d_\ell^2\right\}. \tag{108}$$

Therefore, with probability $1-\alpha/2$

$$|\hat{v}(\boldsymbol{s}) - \mathbb{E}_{\boldsymbol{S}}[\hat{v}(\boldsymbol{s})]| \leq \frac{1}{L-1}\sum_{\ell=1}^{L}\left\{2|\tilde{\tau}_\ell^{\text{pop}} - \hat{\tau}^{\text{pop}}|\epsilon_\ell + \epsilon_\ell^2\right\}. \tag{109}$$

□

Lemma E.1 shows that the errors converge to zero in probability as the sample size $n$ gets large. In particular, $|\hat{v}(\boldsymbol{s}) - \mathbb{E}_{\boldsymbol{S}}[\hat{v}(\boldsymbol{s})]| = o_p(1)$. In practice, the population values $\tilde{\tau}_\ell^{\text{pop}}$ and $\hat{\tau}^{\text{pop}}$ are unknown, but estimates $\hat{\sigma}_\ell$ can be plugged into $\epsilon$ to gauge the magnitude of the bound for a particular set of MC draws $\tilde{y}_{j\ell}$ from the posterior predictive distribution.

# F Other statistical adjustment methods not yet recommended for SEP projects

This section briefly describes other commonly used adjustment methods that are not currently recommended for SEP projects, including calibration (Section F.1) and Multi-level modeling with post-stratification (Section F.2). While these methods are not currently recommended for SEP projects, they may be promising for future investigation.

## F.1 Calibration

Whereas post-stratification re-weights groups defined jointly by multiple variables (e.g., fields that are in Land Resource Region 'M', are predominantly loam, and changed from conventional tillage to no-till), calibration aims to match the marginal distributions. Continuing this example, calibration would weight fields (or sub-fields) so that the proportion of fields in Land Resource Region 'M' is equal to the proportion of total land in the project that is in Land Resource Region 'M', with the same calibrations for the proportion of fields that are predominantly loam and changed from tillage to no-till. However, unlike post-stratification, calibration does not guarantee that the proportion of land with all three of the above attributes would be the same as the proportion of total land in the project with all three attributes (matching marginal distributions does not guarantee matching joint distributions).

Calibration is typically implemented with the raking algorithm [42], though as Zieschang [43] notes, other algorithms can be used. Calibration is also closely related to matching-adjusted indirect comparisons [44], which is used in the health sciences to weight one study cohort to have the same marginal distribution of patient characteristics as another cohort for which only summary statistics are known.

The considerations for which variables to include in the calibration are the same as for post-stratification, and are outlined in Section 5.2.1. Unlike post-stratification, small sample size is not likely to be a challenge for calibration.

Calibration can be useful when only the marginal distributions are known, or when the joint distributions are independent (e.g., the probability a field is predominantly loam is independent of the probability a field changed from tillage to no tillage, and both of these are independent of being in Land Resource Region 'M'). This may not be the case for SEP projects given the relationship between management practices, crop functional groups, and other agricultural considerations, so we do not currently recommend calibration as an adjustment method. However, calibration has been used in agricultural and environmental surveys such as the Bureau of Land Management Rangeland Survey [45], so may be a promising technique for future investigation.

## F.2 Multi-level regression with post-stratification

When post-stratifying on many variables, some adjustment cells may have small sample sizes leading to large variability. This variability can be reduced in two ways: by smoothing the adjustment weights [46] or by using a multilevel model to estimate the cell means. The latter approach is referred to as *multilevel regression with post-stratification* (MRP) [14, 47]. MRP has become a well-established technique in studies of public opinion [48, 49, 50] and health [51, 52, 53], and it is used by the US Census Bureau [54] and the US Centers for Disease Control [55].

Another benefit of MRP is its ability to handle post-strata that are *empty* (i.e., do not have any units that belong to the sample). Post-stratification on many discrete variables can result in many empty post-strata, requiring post-strata to be collapsed (recall Section 5.2.3) using heuristics, manual methods, or expert judgment. If one post-stratifies using a small number of variables with few categories each, the post-strata that result from collapsing cells are easy to interpret because they will be few in number; the list of post-strata could fit onto one page. However, with 5 variables and 5 categories each, there are $5^5 = 3125$ possible post-strata, and even collapsing them into triples requires that a verifier examine a thousand post-strata for their agronomic plausibility, spread across two dozen pages of text.

MRP avoids these issues associated with completely collapsing (or "pooling") cells by *partially pooling*. A multilevel model of cell-means borrows strength among post-strata in an automatic, principled way. For example, suppose there were no samples taken in the post-stratum defined by the triple

$$(\texttt{clay} < 20\%, \texttt{Land Resource Region} = \text{M}, \texttt{tillage intensity reduced}),$$

but there were samples taken in similar cells such as ($20\% < \texttt{clay} < 40\%$, `Land Resource Region` = M, `tillage intensity reduced`). The multilevel model uses information from these similar cells to make predictions in cells with small sample size or even zero samples, in ways that are statistically principled. The estimated population means within each post-stratum are then weighted by the known population counts of the post-strata.

At this time, we view MRP as a promising technique for both reducing absolute bias from missing data and for reducing variance, and for enabling SEP projects to post-stratify on more variables in statistically modern ways. However, while MRP is widely used in social [48, 49, 50] and health sciences [51, 52, 53], we are not aware of examples of MRP being used in agricultural surveys or environmental monitoring. Therefore, we recommend that approaches to MRP in SEP projects be revisited at a future date: once more data are available, prototypes of MRP for SEP projects can be developed. Equipped with these prototypes and the lessons learned in building them, the guiding principles for the use of MRP in SEP projects (such as types of priors to use [56], diagnostic checks, and so on) can then be documented and reviewed by independent experts, thereby enabling verifiers to judge the sound use of MRP in SEP projects.

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